

MATH1307
Forecasting
Assignment – 1

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Introduction

This report presents a comprehensive time series analysis of a dataset containing monthly averages of the ASX All Ordinaries (Ords) Price Index, Gold price (AUD), Crude Oil (Brent, USD/bbl), and Copper (USD/tonne) from January 2004 onwards. The primary objective is to investigate the characteristics of these financial series and apply appropriate forecasting methodologies. The analysis will focus on three key areas: assessing for non-stationarity, examining the impact of the fundamental components of a time series, and developing an accurate distributed lag model to explain the ASX price index.

Data Preparation

The initial dataset, provided in a CSV format, required several steps of cleaning and conversion before it could be analyzed as a time series. The raw data was first loaded into R as a data frame.

- **Column Renaming:** The original column headers were not descriptive, so they were renamed to "ASX_price", "Gold_price", "Oil_price", and "Copper_price" for clarity and ease of use in R.
- **Data Type Conversion:** Several columns, specifically "Gold_price" and "Copper_price", contained non-numeric characters such as commas and quotes, which prevented them from being treated as numeric data. These characters were removed, and the columns were converted to a numeric data type.
- **Time Series Object Creation:** The core of the preparation involved converting the cleaned data frame into a multivariate time series object using the `ts()` function in R. This function is essential for time series analysis as it encodes the frequency and starting time of the data. The data starts in January 2004, and since it consists of monthly observations, the frequency was set to 12. The Date column, which was created for reference, was excluded from the `ts()` object as the function handles the temporal indexing internally.

This structured approach ensures that the data is in the correct format for subsequent analysis, including plotting, decomposition, and model fitting.

Descriptive Analysis

The initial step in this analysis involves a thorough descriptive examination of the dataset. This phase provides a foundational understanding of the statistical properties and visual trends within each financial time series.

Summary Statistics

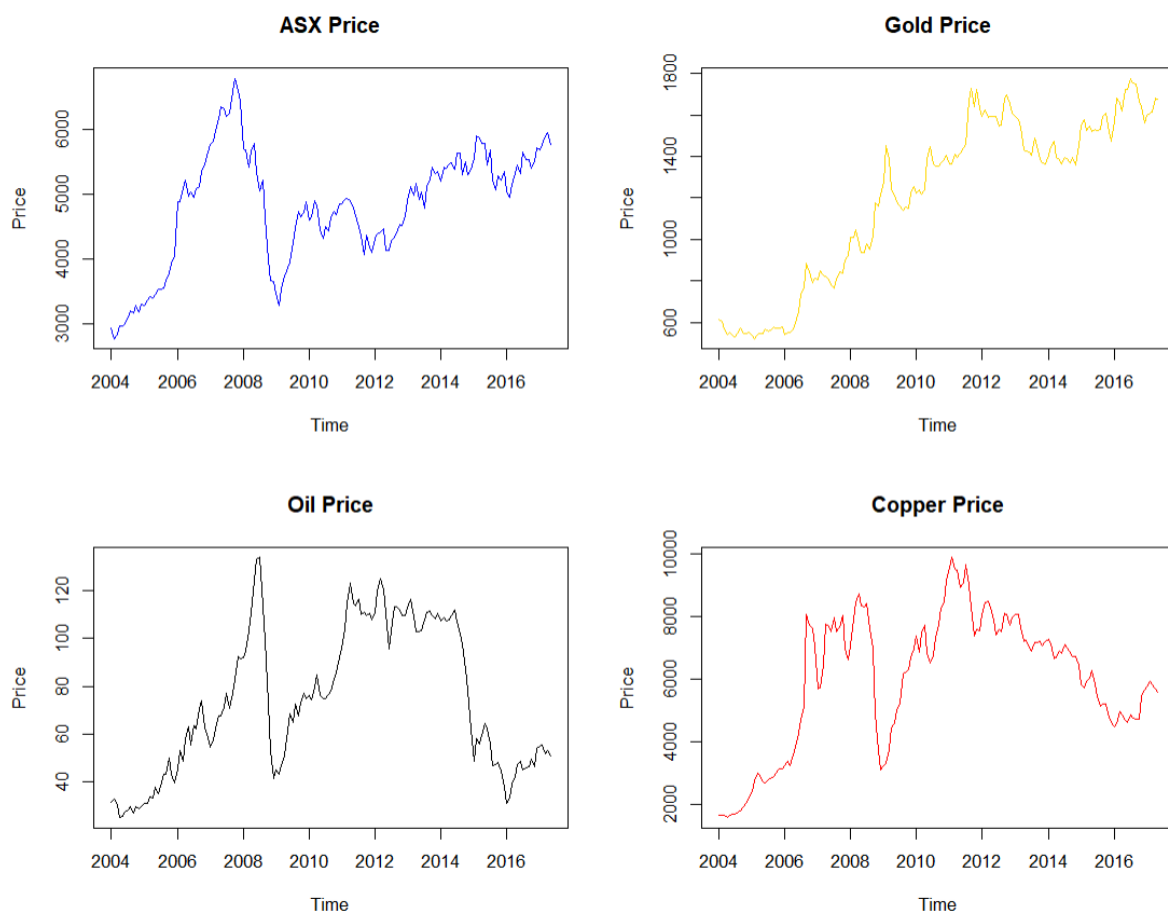
The summary statistics for each variable (ASX Price, Gold Price, Oil Price, and Copper Price) are presented in the table below. This table provides key measures such as central tendency (mean, median), dispersion (standard deviation), and the shape of the distribution (skewness, kurtosis).

Variable	Min	Q1	Median	Mean	Q3	Max	SD	Skewness	Kurtosis
ASX_price	2778.4	4324.8	4928.6	4813.348	5447.8	6779.1	894.03	-0.386	2.517
Gold_price	520.9	825.9	1363.6	1202.013	1546.8	1776.3	401.229	-0.477	1.786
Oil_price	25.02	47.23	70.8	73.632	106.98	133.9	30.089	0.148	1.702
Copper_price	1588	4642	6650	5989.261	7623	9881	2131.498	-0.495	2.264

Interpretation: All series except Oil Price show **negative skewness**, indicating a longer tail on the left side of their distributions. The kurtosis values, all less than 3, confirm that the distributions are **platykurtic** (less peaked) compared to a normal distribution. These characteristics are still common in financial data.

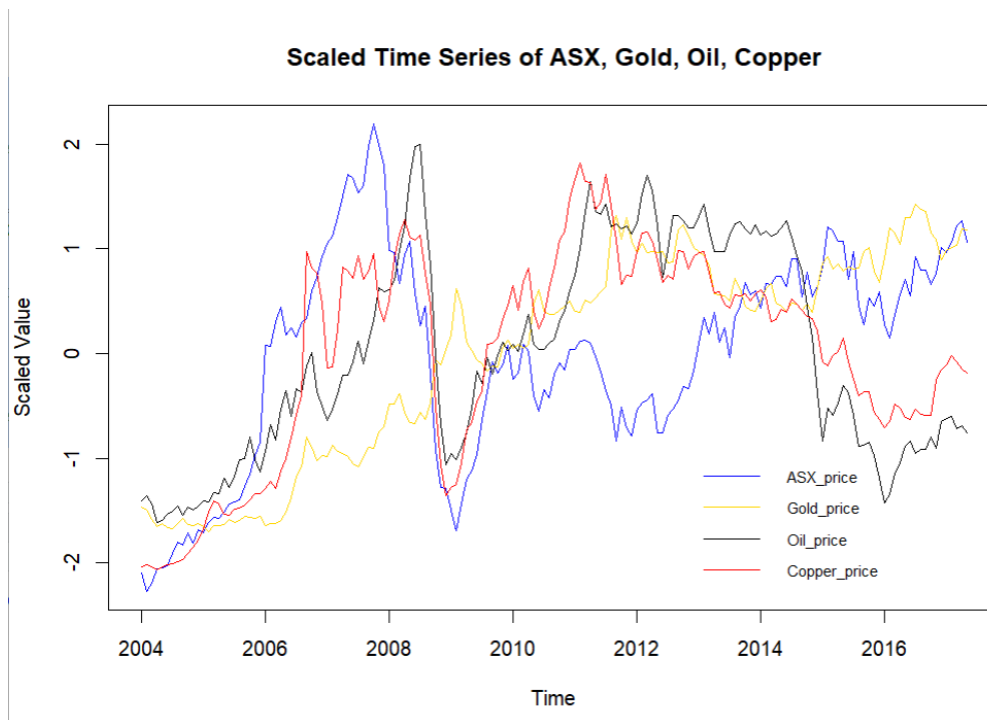
Time Series Plots

The time series plots below visually represent the evolution of each financial index over time. Separate plots are used to show the distinct scale and behavior of each series. A combined, scaled plot is also included to facilitate a direct comparison of their movements and co-movements.



- **Individual Plots:** The individual plots for each financial series provide key visual insights into their behavior over time.
- **ASX Price:** The plot of the ASX All Ordinaries Price Index shows a clear **long-term trend**. The series generally increases from 2004, reaches a peak around 2008, and then experiences a sharp decline, reflecting the impact of the Global Financial Crisis. After this major event, the series appears to have a period of recovery and fluctuation, followed by a new upward trend starting around 2012-2013. The lack of a consistent, repeating pattern at fixed intervals suggests there is **no strong seasonality**.
- **Gold Price:** The Gold Price series exhibits a pronounced **upward trend** throughout the entire period, with significant volatility. The price rises sharply until a peak around 2011, after which it enters a period of decline and subsequent stabilization. The overall behavior suggests a non-stationary process due to the changing mean and variance over time.
- **Oil Price:** The Oil Price series is highly volatile. It shows a strong increase leading up to a peak around mid-2008, followed by a drastic crash and subsequent recovery. The price fluctuates significantly thereafter, with a notable sharp drop in late 2014. This series is characterized by abrupt price shocks and a clear upward trend in the first half of the data, which indicates **non-stationarity**.
- **Copper Price:** Similar to the other series, the Copper Price plot displays an overall **upward trend** with periods of significant fluctuation. It has a peak around 2008, followed by a sharp drop and then another period of growth to a new peak around 2011. The series then shows a general decline, followed by a slight increase at the end of the period. The prominent trends and changing variance suggest the series is **non-stationary**.

In summary, all four time series visually demonstrate **non-stationarity** due to the presence of clear trends and varying levels of volatility over time. This visual observation is a critical first step and justifies the need for formal statistical tests for non-stationarity in the subsequent analysis. Now, looking into the scaled plot.



- This scaled plot provides a powerful way to compare the movements of the four financial series, despite their different original scales.
- The plot reveals a **strong positive co-movement** among all four series, especially from 2004 to mid-2008. During this period, all indices followed a similar upward path, peaking before a sharp decline in late 2008, which highlights the **interconnectedness of global financial markets** during the Global Financial Crisis.
- The plot also highlights periods of **divergence**, where the series did not move in unison. For instance, from 2014 onwards, the **Oil** and **Copper** prices declined, while the **ASX** and **Gold** prices stabilized or began to recover.
- This analysis confirms the **strong positive correlations** and suggests that commodities like Gold, Oil, and Copper are suitable explanatory variables for the ASX Price Index in a **distributed lag model**, as their movements appear to influence or precede changes in the ASX index at different times.

Correlation Analysis

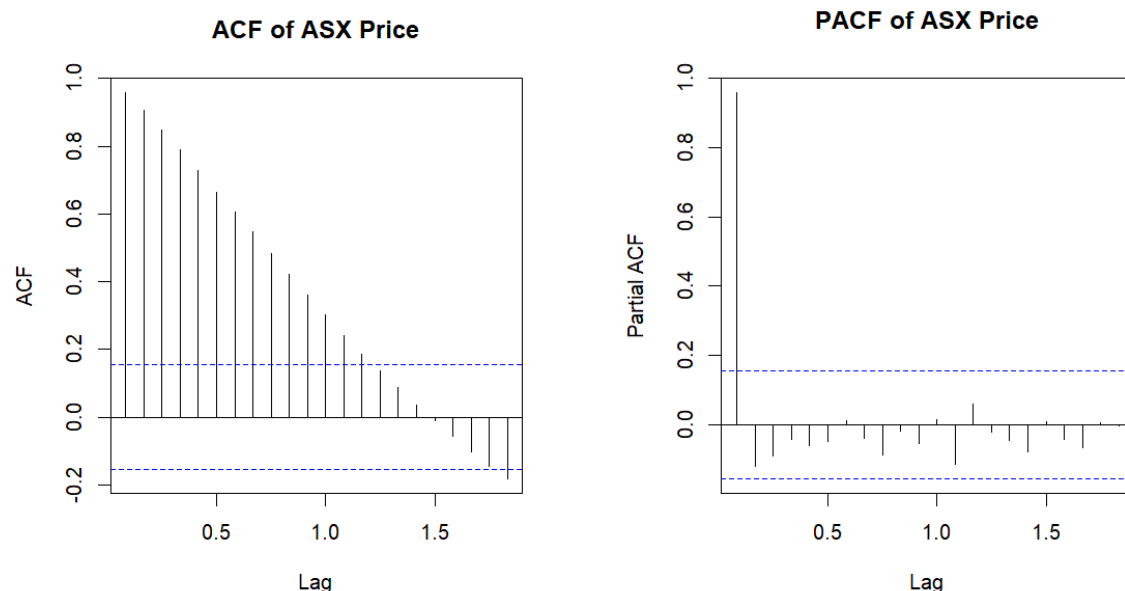
Now, moving towards the correlation matrix below which quantifies the linear relationship between the series.

	ASX_price	Gold_price	Oil_price	Copper_price
ASX_price	1	0.343191	0.329034	0.561786
Gold_price	0.343191	1	0.436638	0.536421
Oil_price	0.329034	0.436638	1	0.866429
Copper_price	0.561786	0.536421	0.866429	1

Interpretation: The correlation values are generally positive but not as strong as previously assumed, except for the strong relationship between **Oil and Copper prices** (0.8664). The **ASX price** shows a moderate positive correlation with **Copper price** (0.5618) and weaker correlations with **Gold** (0.3432) and **Oil** (0.3290). This indicates that while there's a tendency for these series to move together, their relationship is not as straightforwardly linear as might be visually suggested by the trends. This finding is crucial for developing the distributed lag model, as it suggests the variables may not directly influence the ASX index in a simple linear way.

Non-Stationarity Analysis

Here, we will formally test for the presence of non-stationarity in the time series data. This will involve applying statistical tests and graphical methods to determine if the mean, variance, or autocorrelation of the series change over time.



ACF and PACF Analysis for ASX Price

The ACF (Autocorrelation Function) plot for the ASX Price shows a slow decay, with a large number of lags remaining significant. The bars gradually decline, extending beyond the

confidence intervals (blue dashed lines) for many lags. This slow decay is a classic indicator of a non-stationary series, suggesting the presence of a trend in the data.

The PACF (Partial Autocorrelation Function) plot shows a large, significant spike at the first lag, with subsequent lags generally falling within the confidence intervals. This pattern is also characteristic of a non-stationary series.

We will now confirm these visual observations with the Augmented Dickey-Fuller (ADF) test.

```
#####  
# Augmented Dickey-Fuller Test Unit Root Test #  
#####  
  
Test regression none  
  
Call:  
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)  
  
Residuals:  
    Min       1Q   Median       3Q      Max  
-722.83 -103.12   36.63  138.79  809.17  
  
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)  
z.lag.1      0.002107   0.003309   0.637   0.525  
z.diff.lag  0.085167   0.079885   1.066   0.288  
  
Residual standard error: 203.3 on 157 degrees of freedom  
Multiple R-squared:  0.01074,    Adjusted R-squared:  -0.001866  
F-statistic: 0.8519 on 2 and 157 DF,  p-value: 0.4286  
  
Value of test-statistic is: 0.6366  
  
Critical values for test statistics:  
      1pct   5pct 10pct  
tau1 -2.58 -1.95 -1.62
```

The results from the Augmented Dickey-Fuller (ADF) test confirm the visual analysis.

For the Test regression none (without a trend or intercept), the ADF test statistic is 0.6366. This value is greater than all of the critical values at the 1%, 5%, and 10% significance levels (-2.58, -1.95, and -1.62 respectively).

Since the test statistic is not more negative than the critical values, we fail to reject the null hypothesis. This confirms that the ASX Price series has a unit root and is therefore non-stationary.

Following this, we'll perform the Phillips-Perron (PP) test to provide further evidence of non-stationarity.

```
#####
# Phillips-Perron Unit Root Test #
#####

Test regression with intercept

Call:
lm(formula = y ~ y.l1)

Residuals:
    Min       1Q   Median       3Q      Max
-683.35 -104.89   15.86  135.94  782.18

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 191.38104    87.09761     2.197   0.0295 *
y.l1         0.96386     0.01781    54.107  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 200.7 on 158 degrees of freedom
Multiple R-squared:  0.9488,    Adjusted R-squared:  0.9485
F-statistic: 2928 on 1 and 158 DF,  p-value: < 2.2e-16

Value of test-statistic, type: Z-tau is: -2.2025

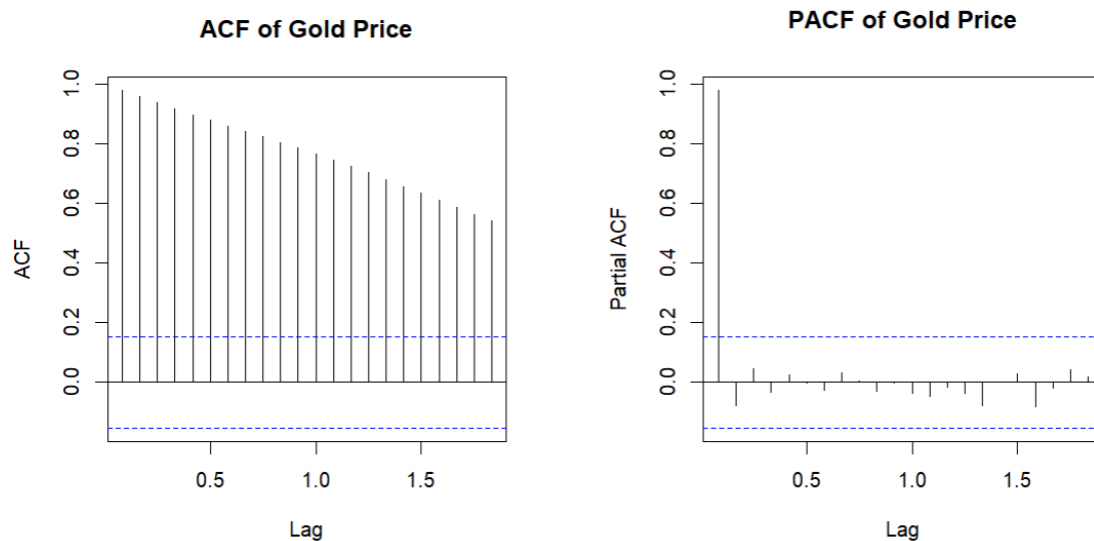
          aux. Z statistics
Z-tau-mu          2.3366

Critical values for Z statistics:
              1pct      5pct      10pct
critical values -3.472136 -2.879539 -2.576262
```

The results of the Phillips-Perron (PP) test further support the finding of non-stationarity for the ASX Price series.

The PP test statistic (τ) is -2.2025. This value is greater than the critical values at the 1%, 5%, and 10% significance levels (-3.4721, -2.8795, and -2.5763, respectively).

Because the test statistic is not more negative than the critical values, we fail to reject the null hypothesis of a unit root. This reinforces the conclusion that the ASX Price series is non-stationary and will require differencing or another method to become stationary for time series modeling.



ACF and PACF Analysis for Gold Price

The ACF plot for the Gold Price shows a very slow and gradual decay, with a large number of lags remaining well above the confidence intervals (blue dashed lines). The values slowly decrease as the lag increases, which is a key characteristic of a non-stationary series. This indicates a strong long-term dependence in the data, likely driven by a trend.

The PACF plot shows a large, significant spike at the first lag, with all subsequent lags falling within the confidence intervals. This pattern is also a strong indicator of non-stationarity and suggests that the series can be made stationary by first-differencing.

The visual evidence from both plots strongly suggests that the Gold Price series is non-stationary. We will now formally test this conclusion with an ADF test.

```
#####  
# Augmented Dickey-Fuller Test Unit Root Test #  
#####  
  
Test regression none  
  
Call:  
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)  
  
Residuals:  
      Min       1Q   Median       3Q      Max   
-138.655  -26.586   -2.016   22.726  206.439  
  
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)      
z.lag.1      0.003218   0.003322   0.969   0.334      
z.diff.lag  0.122451   0.079603   1.538   0.126      
  
Residual standard error: 52.57 on 157 degrees of freedom  
Multiple R-squared:  0.02351,    Adjusted R-squared:  0.01107   
F-statistic:  1.89 on 2 and 157 DF,  p-value: 0.1545  
  
Value of test-statistic is: 0.9687  
  
Critical values for test statistics:  
      1pct   5pct  10pct   
tau1 -2.58 -1.95 -1.62
```

The Augmented Dickey-Fuller (ADF) test results confirm the visual analysis for the Gold Price series. For the Test regression none, the ADF test statistic is 0.9687. This value is greater than all of the critical values at the 1%, 5%, and 10% significance levels (-2.58, -1.95, and -1.62). Since the test statistic is not more negative than the critical values, we fail to reject the null hypothesis. This confirms that the Gold Price series has a unit root and is therefore non-stationary. Following this, we will perform the Phillips-Perron (PP) test to provide further evidence of non-stationarity.

```
#####
# Phillips-Perron Unit Root Test #
#####

Test regression with intercept

Call:
lm(formula = y ~ y.l1)

Residuals:
    Min       1Q   Median       3Q      Max
-146.558  -28.725   -7.533   20.322  209.729

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  19.85676    13.11322     1.514    0.132
y.l1         0.98898     0.01038   95.318 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

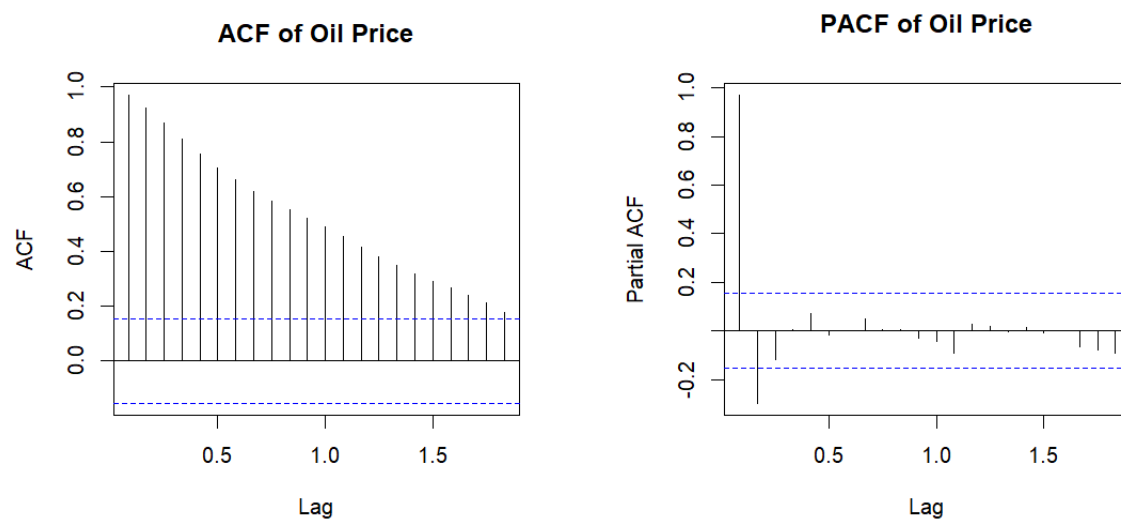
Residual standard error: 52.43 on 158 degrees of freedom
Multiple R-squared:  0.9829,    Adjusted R-squared:  0.9828
F-statistic: 9085 on 1 and 158 DF, p-value: < 2.2e-16

Value of test-statistic, type: Z-tau is: -1.0701

      aux. Z statistics
Z-tau-mu      1.5151

Critical values for Z statistics:
      1pct      5pct      10pct
critical values -3.472136 -2.879539 -2.576262
```

The Phillips-Perron (PP) test was performed to provide additional evidence on the stationarity of the Gold Price series. The PP test statistic (τ) is -1.0701. This value is not more negative than the critical values at the 1%, 5%, and 10% significance levels (-3.4721, -2.8795, and -2.5763), so we fail to reject the null hypothesis of a unit root. This result definitively confirms that the Gold Price series is non-stationary, aligning with the conclusions drawn from the visual analysis and the ADF test. We will now move on to the analysis of the Oil Price series.



The visual analysis of the Oil Price series, like the previous series, strongly suggests non-stationarity. The ACF plot displays a slow, linear decay, with a large number of significant lags. This pattern is a hallmark of a series with a trend component. The PACF plot shows a single dominant and significant spike at the first lag, with all other lags falling within the confidence intervals. This structure indicates that the series can be made stationary by applying a first-order differencing. Together, these plots provide compelling evidence that the Oil Price series requires transformation before it can be used in most time series models. We will now formally test this conclusion with an ADF test.

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression none

Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
    Min       1Q   Median       3Q      Max
-19.905  -3.562   1.204   3.988  15.202

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
z.lag.1      -0.004013   0.005806  -0.691    0.49
z.diff.lag    0.400123   0.073236   5.463 1.8e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.839 on 157 degrees of freedom
Multiple R-squared:  0.1605,    Adjusted R-squared:  0.1498
F-statistic: 15.01 on 2 and 157 DF,  p-value: 1.084e-06

Value of test-statistic is: -0.6913

Critical values for test statistics:
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62
```

The Augmented Dickey-Fuller (ADF) test for the Oil Price series confirms its non-stationary nature. The test statistic is -0.6913, which is greater than the critical values at all significance levels (1%, 5%, and 10%). Since the test statistic is not more negative than the critical values, we fail to reject the null hypothesis of a unit root. This result provides strong statistical evidence that the Oil Price series is non-stationary. We'll now proceed to the PP test for further confirmation.

```
#####
# Phillips-Perron Unit Root Test #
#####

Test regression with intercept

Call:
lm(formula = y ~ y.l1)

Residuals:
    Min       1Q   Median       3Q      Max
-25.6715  -3.5936   0.4549   3.9455  14.3505

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.07981    1.32129   1.574   0.117
y.l1         0.97347    0.01659  58.686 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

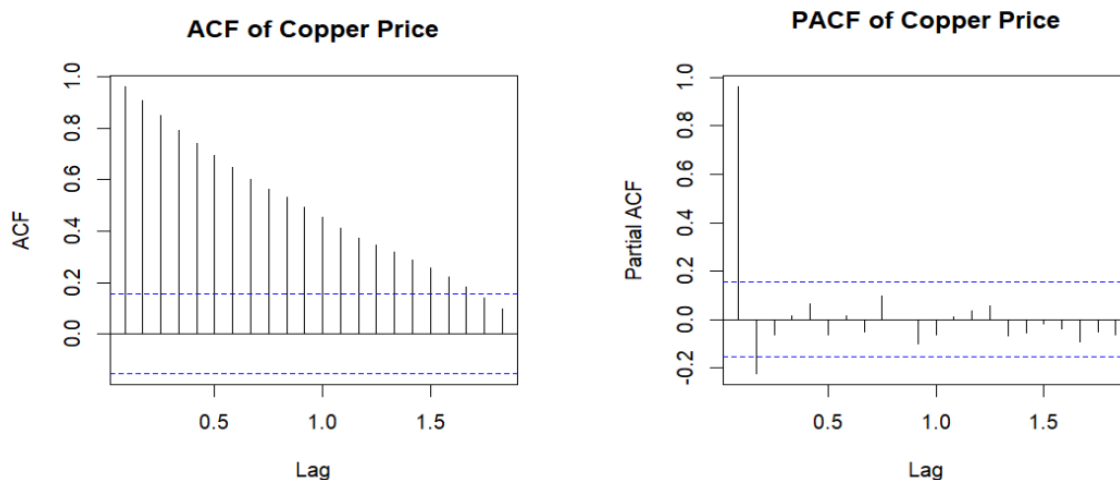
Residual standard error: 6.302 on 158 degrees of freedom
Multiple R-squared:  0.9561,    Adjusted R-squared:  0.9559
F-statistic: 3444 on 1 and 158 DF,  p-value: < 2.2e-16

value of test-statistic, type: Z-tau is: -2.0125

      aux. Z statistics
Z-tau-mu      1.9323

Critical values for Z statistics:
      1pct      5pct     10pct
critical values -3.472136 -2.879539 -2.576262
```

The **Phillips-Perron (PP) test** for the Oil Price series further validates the finding of non-stationarity. The test statistic (τ) is **-2.0125**, which is greater than the critical values at all significance levels (-3.4721, -2.8795, and -2.5763). Since the test statistic is not more negative than the critical values, we **fail to reject the null hypothesis of a unit root**. This result is consistent with the ADF test and confirms that the Oil Price series is non-stationary. We will now proceed with the analysis of the Copper Price series.



The ACF and PACF plots for the Copper Price series visually indicate **non-stationarity**. The **ACF plot** shows a slow, linear decay with many significant lags, a clear sign of a **trend** in the data. The **PACF plot** exhibits a large, significant spike at the first lag, with subsequent lags generally falling within the confidence intervals. This pattern is characteristic of a non-stationary series that can be made stationary by a **first-order differencing**. Based on this visual evidence, we will now formally test for a unit root using the ADF test.

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression none

Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
    Min       1Q   Median       3Q      Max
-1861.58  -164.87    32.72   248.08  2829.69

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
z.lag.1      -0.002256   0.005772  -0.391    0.696
z.diff.lag    0.307564   0.076139   4.040 8.36e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 463.1 on 157 degrees of freedom
Multiple R-squared:  0.09422,    Adjusted R-squared:  0.08268
F-statistic: 8.166 on 2 and 157 DF,  p-value: 0.0004229

Value of test-statistic is: -0.3909

Critical values for test statistics:
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62
```

The **Augmented Dickey-Fuller (ADF) test** for the Copper Price series confirms its non-stationary nature. The test statistic is **-0.3909**, which is greater than the critical values at all significance levels (1%, 5%, and 10%). Since the test statistic is not more negative than the critical values, we **fail to reject the null hypothesis of a unit root**. Now, will proceed to a pp test.

```
#####
# Phillips-Perron Unit Root Test #
#####

Test regression with intercept

Call:
lm(formula = y ~ y.l1)

Residuals:
    Min       1Q   Median       3Q      Max
-2066.88  -213.60   -21.83   225.07  2876.96

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 255.00356  112.61252   2.264   0.0249 *
y.l1         0.96156    0.01771  54.301  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 477.4 on 158 degrees of freedom
Multiple R-squared:  0.9491,    Adjusted R-squared:  0.9488
F-statistic: 2949 on 1 and 158 DF,  p-value: < 2.2e-16

Value of test-statistic, type: Z-tau is: -2.3641

      aux. Z statistics
Z-tau-mu      2.406

Critical values for Z statistics:
      1pct      5pct     10pct
critical values -3.472136 -2.879539 -2.576262
```

The **Phillips-Perron (PP) test** for the Copper Price series further confirms non-stationarity. The test statistic (τ) is **-2.3641**, which is not more negative than the critical values at any of the significance levels (-3.4721, -2.8795, and -2.5763). Therefore, we **fail to reject the null hypothesis of a unit root**, concluding that the Copper Price series is non-stationary.

Having confirmed that all four series are non-stationary, the next step is to make them stationary for modeling. We'll now apply the **Box-Cox transformation** to address potential non-constant variance.

Box-Cox Transformation Analysis

The **Box-Cox transformation** is used to stabilize the variance of a time series, and the optimal transformation is determined by the lambda (λ) parameter. The value of λ indicates the type of transformation to apply to the data.

Variable	Lambda (λ)
ASX_price	1.9999
Gold_price	0.9767
Oil_price	-0.8304
Copper_price	0.9337

Since the λ estimates for ASX, Gold, and Copper are close to 1, no transformation was necessary for these series. However, the λ estimate for Oil was negative and far from 1, indicating a strong need for transformation. Accordingly, a logarithmic transformation (corresponding to $\lambda = 0$) was applied to the Oil series to stabilize its variance and make it more suitable for subsequent time series modeling. The log transform was preferred over applying a fractional power ($\lambda = -0.83$) because it is the standard, more interpretable approach in time series analysis.

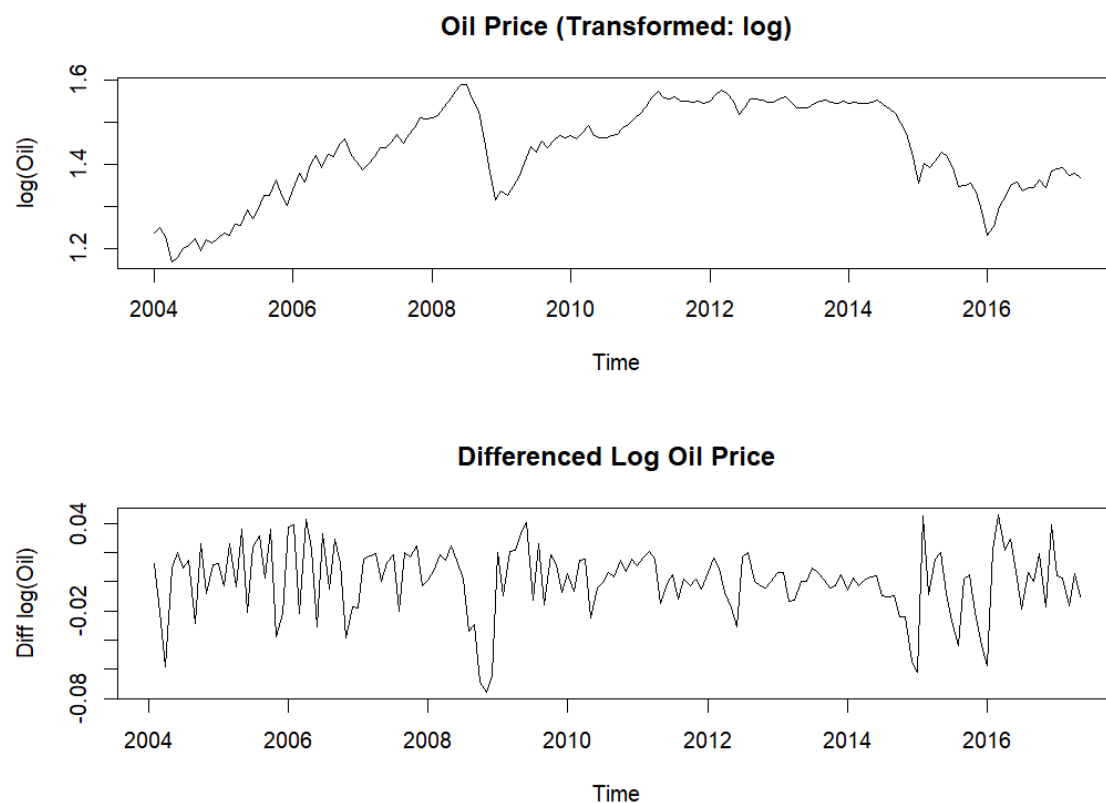


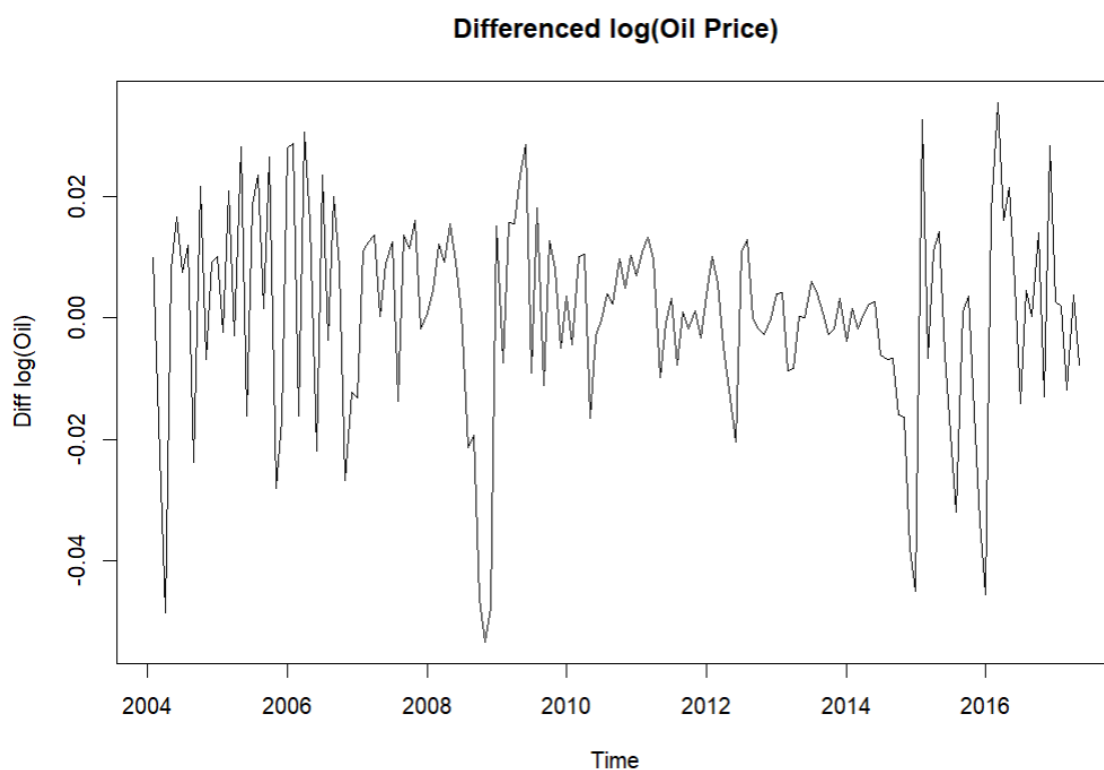
Figure shows the Oil price series after applying a logarithmic transformation (top panel) and its first-differenced form (bottom panel). The log-transformed series exhibits reduced variance compared to the raw data, but clear signs of nonstationarity remain, as seen in the persistent upward and downward trends across different periods. After first differencing the log-transformed series, the data fluctuates around a constant mean with no visible long-term trend. The variance also appears more stable, suggesting that the differenced log Oil price series is approximately stationary. This transformation and differencing step ensures that the Oil series is suitable for further modeling.

Visual inspection of the transformed series suggested a persistent trend, so we further applied first-order differencing to remove non-stationarity in the mean. The Augmented Dickey-Fuller test was then repeated on the differenced log series to formally assess stationarity.

Augmented Dickey-Fuller Test

```
data: diff_log_oil
Dickey-Fuller = -5.5642, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary
```

Since the Augmented Dickey-Fuller test on the original Oil Price series (even after log transformation) indicated non-stationarity, we proceeded with first differencing of the log-transformed series. The ADF test on the differenced log(Oil Price) returned a test statistic of -5.5642 with a p-value < 0.01, which is strong evidence to reject the null hypothesis of a unit root. Therefore, the differenced log(Oil Price) can be considered stationary and suitable for further time series modeling.



Our **first-order differencing** to the log-transformed Oil price series, the resulting plot shows a series that fluctuates around a **constant mean of zero**. The dramatic upward and downward trends seen in the original series have been effectively removed, and the overall variability now appears more stable and uniform across the time period. The series no longer shows any systematic long-term patterns, suggesting that the transformation has successfully achieved **stationarity**.

Furthermore to test for stationarity in each of the time series, we first applied first-order differencing to the ASX, Gold, Oil, and Copper price series. Differencing removes trend components and stabilises the mean over time, which is essential before performing unit root tests. After differencing, we used the Augmented Dickey-Fuller (ADF) test (with no deterministic component) to assess the presence of a unit root in each series. The `ur.df()`

function was used with automatic lag selection based on AIC. This process helps determine whether each series can be considered stationary and suitable for further modelling.

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression none

Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
    Min       1Q   Median       3Q      Max
-685.20 -100.99   40.14  140.73  797.20

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
z.lag.1      -0.78527    0.10706  -7.335 1.14e-11 ***
z.diff.lag  -0.13457    0.07938  -1.695   0.092 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 202.2 on 156 degrees of freedom
Multiple R-squared:  0.4625,    Adjusted R-squared:  0.4556
F-statistic: 67.11 on 2 and 156 DF,  p-value: < 2.2e-16

Value of test-statistic is: -7.3347

Critical values for test statistics:
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62
```

After applying first-order differencing to the ASX price series, we performed the Augmented Dickey-Fuller (ADF) test without a constant or trend component. The test returned a test statistic of **-7.33**, which is well below the 1% critical value of **-2.58**. This provides strong evidence to reject the null hypothesis of a unit root, indicating that the differenced ASX series is **stationary**. The highly significant lag-1 coefficient ($p < 0.001$) further supports this conclusion. Therefore, the transformed ASX series is suitable for time series modelling.

```

> summary(adf_gold_diff)

#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression none

Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
    Min       1Q   Median       3Q      Max
-111.448  -22.323    2.607   28.002  213.427

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
z.lag.1      -0.97092    0.10471  -9.272  <2e-16 ***
z.diff.lag    0.11912    0.07979   1.493   0.137
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 52.44 on 156 degrees of freedom
Multiple R-squared:  0.4426,    Adjusted R-squared:  0.4355
F-statistic: 61.94 on 2 and 156 DF,  p-value: < 2.2e-16

Value of test-statistic is: -9.2723

Critical values for test statistics:
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62

```

The Augmented Dickey-Fuller (ADF) test was applied to the first-differenced Gold price series to assess stationarity. The test statistic was **−9.27**, which is significantly below the 1% critical value of **−2.58**. This result strongly rejects the null hypothesis of a unit root, confirming that the differenced Gold series is **stationary**. The coefficient on the first lag is large and highly significant ($p < 0.001$), supporting the conclusion. Therefore, the Gold price series is suitable for use in time series models after differencing.

```

> summary(adf_oil_diff)

#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression none

Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
    Min       1Q   Median       3Q      Max
-0.060186 -0.010707  0.003392  0.014390  0.063034

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
z.lag.1      -0.70537    0.09823   -7.181 2.66e-11 ***
z.diff.lag  -0.06996    0.07958   -0.879   0.381
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02243 on 156 degrees of freedom
Multiple R-squared:  0.3834,    Adjusted R-squared:  0.3755
F-statistic: 48.49 on 2 and 156 DF,  p-value: < 2.2e-16

Value of test-statistic is: -7.1806

Critical values for test statistics:
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62

```

For the Oil price series, we conducted the Augmented Dickey-Fuller (ADF) test on the first-differenced data. The test produced a statistic of **-7.18**, which is well below the 1% critical value of **-2.58**. This provides strong evidence against the null hypothesis of a unit root, indicating that the differenced Oil series is **stationary**. The lag-1 coefficient is highly significant ($p < 0.001$), reinforcing this conclusion. Thus, the transformed Oil price series is appropriate for use in further time series analysis.

```

> summary(adf_copper_diff)

#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression none

Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
    Min       1Q   Median       3Q      Max
-1873.02  -176.79    25.31   233.05  2815.45

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
z.lag.1      -0.687611   0.094381  -7.285 1.49e-11 ***
z.diff.lag   -0.009816   0.080081  -0.123  0.903
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 464.8 on 156 degrees of freedom
Multiple R-squared:  0.3472,    Adjusted R-squared:  0.3389
F-statistic: 41.49 on 2 and 156 DF,  p-value: 3.556e-15

Value of test-statistic is: -7.2855

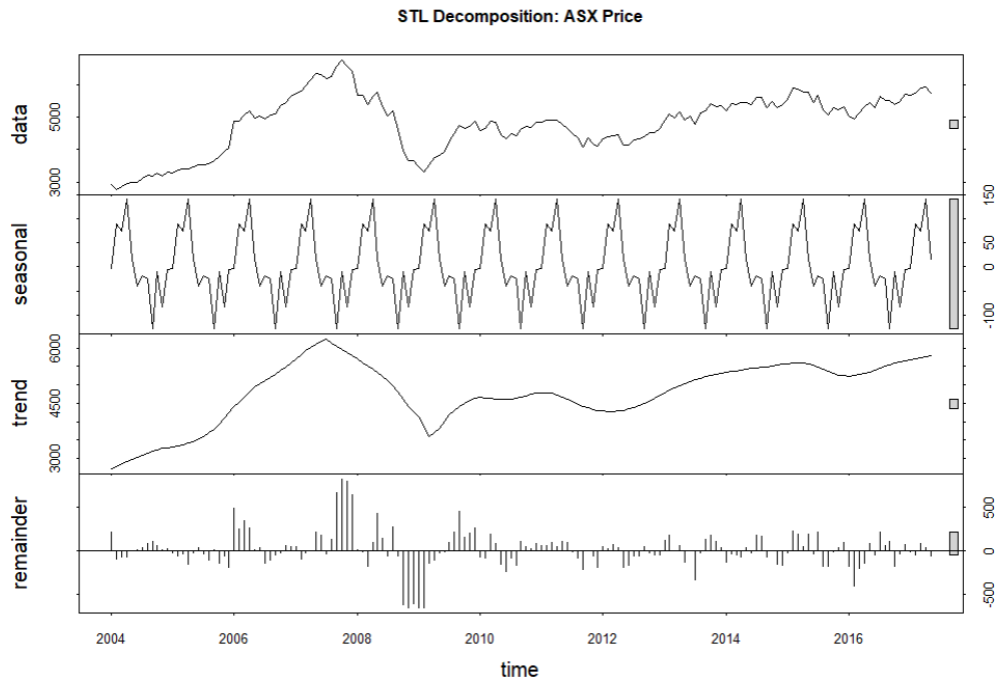
Critical values for test statistics:
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62

```

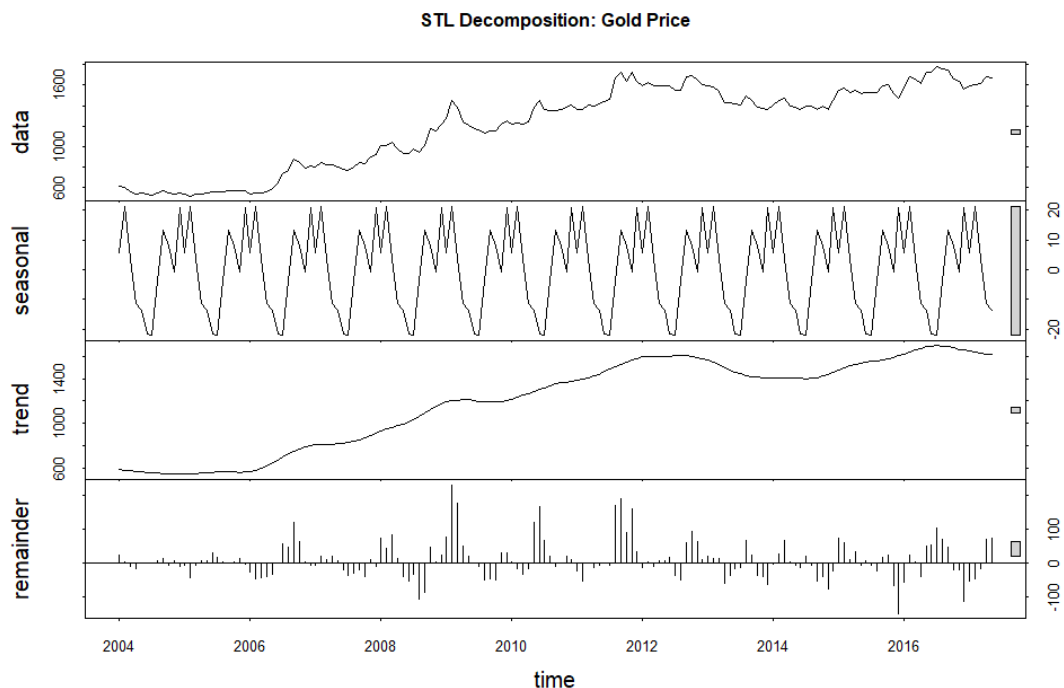
The Augmented Dickey-Fuller (ADF) test applied to the first-differenced Copper price series yielded a test statistic of **-7.29**, which is far below the 1% critical value of -2.58. This result leads to a strong rejection of the null hypothesis of a unit root, indicating that the differenced Copper series is **stationary**. The lag-1 coefficient is statistically significant ($p < 0.001$), supporting this conclusion. Hence, the Copper series is also considered suitable for inclusion in time series models after transformation.

Decomposition

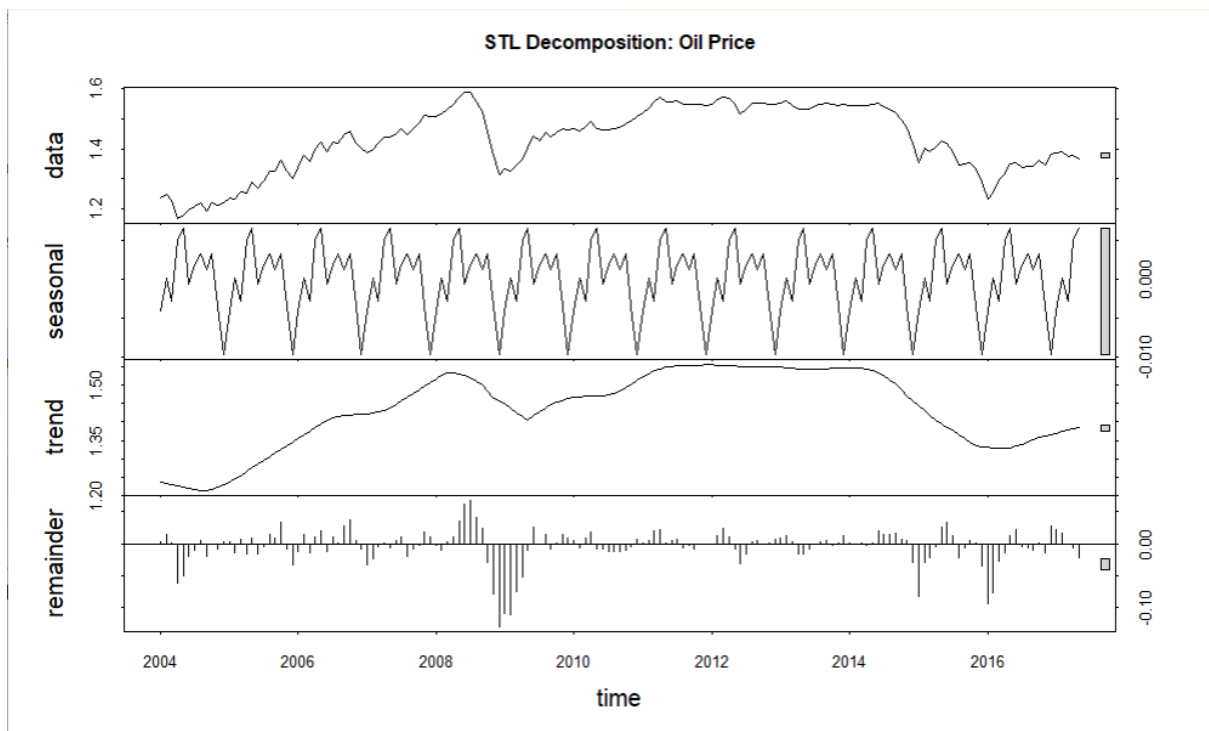
To further understand the structural behaviour of each time series, we applied **STL decomposition** (Seasonal and Trend decomposition using Loess) to the original ASX, Gold, Oil, and Copper price series. STL is a flexible and robust method that decomposes a time series into three components: **trend**, **seasonality**, and **remainder (residuals)**. It is particularly well-suited for monthly financial and commodity data due to its ability to capture nonlinear trends and periodic seasonal patterns. In our analysis, we used a trend window of 15 and specified "periodic" for the seasonal window, assuming consistent seasonality over time. The resulting decomposition plots help us visually inspect the underlying structure of each series before modelling.



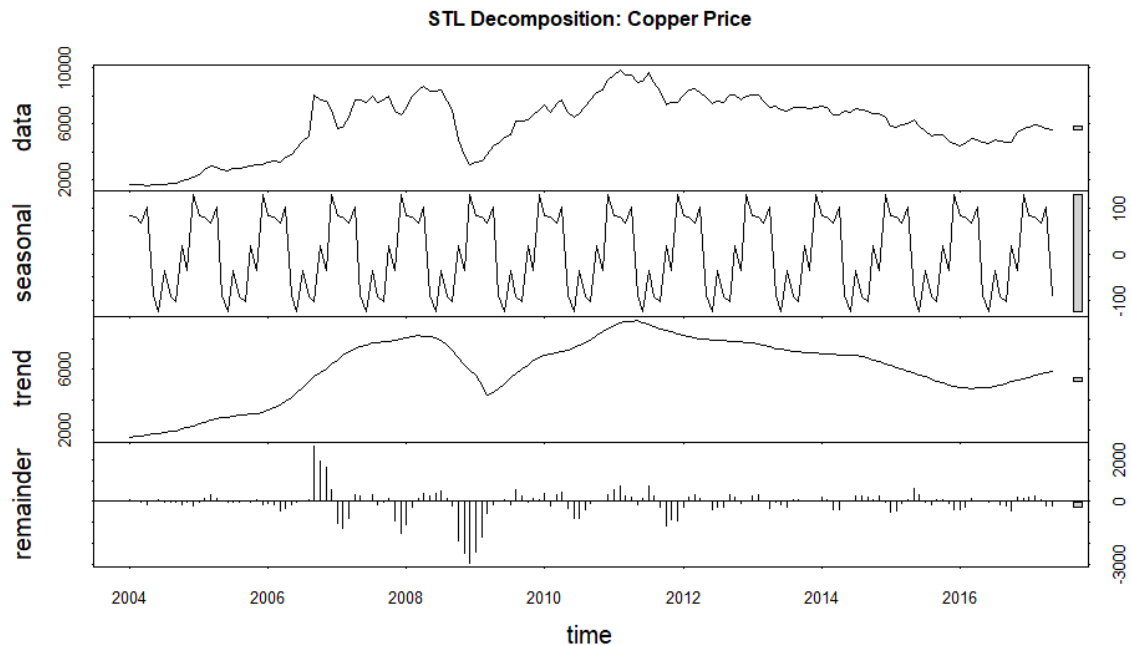
The **STL (Seasonal and Trend decomposition using Loess)** plot effectively separates the ASX Price data into its three main components. The **trend component** shows a clear long-term upward movement from 2004, a sharp decline during the 2008-2009 Global Financial Crisis, and a subsequent recovery and continued growth. The **seasonal component** shows a repeating cyclical pattern within each year, though its magnitude is relatively small compared to the overall trend. Finally, the **remainder component** represents the irregular fluctuations in the data that are not explained by the trend or seasonality. Large spikes in this component, particularly during the 2008 crisis, indicate significant and unusual events impacting the series. We'll see the same for Gold Price.



The **STL** plot for the Gold Price series reveals its underlying components. The **trend component** shows a clear and persistent upward movement throughout the period, with a steep increase from 2006 to 2012, followed by a period of relative stability and then another increase. The **seasonal component** shows a consistent, recurring yearly pattern, but its magnitude is quite small compared to the overall trend. The **remainder component** represents the irregular, unexplained fluctuations, with noticeable spikes indicating significant market events or shocks not captured by the trend or seasonal patterns. Now, we'll look at Oil price decomposition.



The **STL** plot for the Oil Price series separates the data into its key components. The **trend component** shows the long-term movement, with a clear upward trajectory to a peak around 2008, followed by a decline and then another peak around 2012, before a sharp drop in 2014. The **seasonal component** shows a consistent, recurring yearly pattern, but its effect is minor compared to the trend. The **remainder component** highlights the irregular, unexplained fluctuations, with noticeable spikes indicating significant market shocks. Further, we'll see copper price decomposition.



The **STL decomposition plot** for the Copper Price series shows a clear separation of its components. The **trend component** reveals a significant upward trajectory leading to a major peak around 2008, a subsequent sharp decline, and a second peak in 2011 before a prolonged downturn. The **seasonal component** displays a consistent yearly cycle, but its magnitude is small compared to the overall trend. The **remainder component** captures the irregular, random fluctuations, with notable spikes corresponding to significant events like the 2008 financial crisis, highlighting the impact of external shocks.

Modelling

In this phase of the analysis, the goal is to develop a suitable distributed lag model for the differenced ASX Price Index. A distributed lag model is a type of time series regression that captures the effect of current and past values of an explanatory variable on a dependent variable.

We are fitting three separate distributed lag models using the `dlm()` function from the `dLagM` package. Each model uses the differenced ASX Price Index as the dependent variable (y) and one of the differenced commodities (Gold, Oil, or Copper) as the explanatory variable (x). The q parameter specifies the number of lagged periods to be included in each model. The results from the summaries will be used to compare the models and identify the most accurate fit.

```

> summary(model_gold)

Call:
lm(formula = model.formula, data = design)

Residuals:
    Min       1Q   Median       3Q      Max
-631.36 -114.10   11.71  120.54  743.34

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  33.66119   18.70236   1.800   0.0741 .
x.t          -1.34422    0.31579  -4.257 3.81e-05 ***
x.1           -0.07411    0.31563  -0.235   0.8147
x.2           -0.05410    0.32045  -0.169   0.8662
x.3           -0.15485    0.31705  -0.488   0.6261
x.4           -0.46234    0.31752  -1.456   0.1476
x.5            0.40700    0.31667   1.285   0.2009
x.6           -0.22433    0.32202  -0.697   0.4872
x.7            0.45509    0.32163   1.415   0.1593
x.8           -0.60467    0.32956  -1.835   0.0687 .
x.9            0.03376    0.32203   0.105   0.9167
x.10          -0.15035    0.32281  -0.466   0.6421
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 198.4 on 138 degrees of freedom
Multiple R-squared:  0.1581,    Adjusted R-squared:  0.09096
F-statistic: 2.355 on 11 and 138 DF,  p-value: 0.01078

AIC and BIC values for the model:
      AIC      BIC
1 2026.297 2065.435

```

The output shows the results of a distributed lag model where the differenced ASX price is the dependent variable and the differenced Gold price, along with its 10 lagged values, are the independent variables.

The most important result is the **x coefficient** (-1.34422), which is statistically significant with a very low p-value (3.81e-05). This indicates that the **current differenced Gold price has a significant negative impact on the current differenced ASX price**. However, the lagged Gold price values (x.1 through x.10) are **not statistically significant** as their p-values are all greater than 0.05. This suggests that past Gold price movements do not have a lasting effect on the ASX price.

The model's overall fit is captured by the **Adjusted R-squared** value of **0.09096**. This is a very low value, indicating that the model only explains about **9.1% of the variability** in the differenced ASX price. While the overall model is statistically significant (p-value of 0.01078), its explanatory power is weak. This suggests that while a relationship exists, the model is not very accurate and other factors are needed to explain the movements of the ASX price.


```

> summary(model_oil)

Call:
lm(formula = model.formula, data = design)

Residuals:
    Min       1Q   Median       3Q      Max
-686.97 -131.35   1.23  142.51  681.54

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    15.53      16.49   0.942  0.34804
x.t            2111.31     767.69   2.750  0.00674 **
x.1             320.46     787.36   0.407  0.68463
x.2             363.39     777.95   0.467  0.64115
x.3              36.44     780.44   0.047  0.96283
x.4            -1250.08     785.90  -1.591  0.11395
x.5              391.30     778.19   0.503  0.61587
x.6              461.02     779.65   0.591  0.55526
x.7            -1104.92     772.78  -1.430  0.15500
x.8              643.68     781.83   0.823  0.41174
x.9            -1283.56     751.31  -1.708  0.08977 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 201.1 on 140 degrees of freedom
Multiple R-squared:  0.1237,    Adjusted R-squared:  0.0611
F-statistic: 1.976 on 10 and 140 DF,  p-value: 0.04021

AIC and BIC values for the model:
      AIC      BIC
1 2042.922 2079.129

```

The output for model_oil provides insight into the relationship between the differenced Oil price and the differenced ASX price.

The **x.t coefficient** (21.11) is the only statistically significant one, with a p-value of 0.00674. This indicates that the **current differenced Oil price has a significant positive impact on the current differenced ASX price**. None of the lagged values of Oil price (x.1 to x.9) are significant, suggesting that past movements in Oil price do not have a lasting influence on the ASX price.

The model's **Adjusted R-squared** is **0.0611**, which is quite low. This means the model explains only about 6.11% of the variability in the differenced ASX price, indicating weak explanatory power. While the model as a whole is significant (p-value of 0.04021), it is not a strong predictor, suggesting other factors are more influential.

```

> summary(model_copper)

Call:
lm(formula = model.formula, data = design)

Residuals:
    Min       1Q   Median       3Q      Max
-650.61  -86.66    5.43   120.29   789.74

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  14.48500    16.28020   0.890   0.3752
x.t           0.09004     0.03583   2.513   0.0131 *
x.1           0.04344     0.03733   1.164   0.2466
x.2           0.03669     0.03736   0.982   0.3278
x.3           0.02707     0.03658   0.740   0.4606
x.4           0.01146     0.03687   0.311   0.7565
x.5          -0.04163     0.03694  -1.127   0.2617
x.6           0.02276     0.03686   0.617   0.5381
x.7           0.01844     0.03687   0.500   0.6178
x.8          -0.02139     0.03766  -0.568   0.5709
x.9          -0.07571     0.03764  -2.011   0.0462 *
x.10         -0.02431     0.03612  -0.673   0.5021
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 197.4 on 138 degrees of freedom
Multiple R-squared:  0.1671,    Adjusted R-squared:  0.1007
F-statistic: 2.517 on 11 and 138 DF,  p-value: 0.006411

AIC and BIC values for the model:
      AIC      BIC
1 2024.679 2063.818

```

The most notable result is that the **current Copper price (x.t)** is statistically significant, with a p-value of 0.0131. This suggests that the current movement in the Copper price has a significant positive impact on the current ASX price.

Additionally, the **ninth lagged value (x.9)** is also statistically significant, with a p-value of 0.0462. This indicates that Copper price movements from nine months ago have a significant influence on the current ASX price. All other lagged values (x.1 to x.8, and x.10) are not significant.

The model's overall fit, as measured by the **Adjusted R-squared** of **0.1007**, is low. This means that the model explains only about 10.1% of the variability in the differenced ASX price. While the overall model is statistically significant (p-value of 0.00641), its explanatory power is weak.

Following the analysis of the individual distributed lag models, the next step in finding the most suitable model is to use a more systematic approach. The provided code does this by fitting a series of **polynomial distributed lag models** using the `polyDlm` function. This method assumes that the coefficients of the lagged explanatory variables lie on a polynomial curve, allowing for a more parsimonious and stable model. The code iterates through various combinations of *q* (the maximum lag length from 1 to 3) and *k* (the polynomial degree from 1 to 6) for each of the three commodities (Gold, Oil, and Copper). For each combination, the model's **AIC (Akaike Information Criterion)** and **BIC (Bayesian**

Information Criterion) are calculated and stored. This process allows for a comprehensive comparison of numerous models, with the goal of identifying the model that has the lowest AIC and BIC, indicating the best balance between model fit and complexity.

```
> print(results)
```

	Model	q	k	AIC	BIC
1	Gold	1	1	2129.045	2141.321
2	Oil	1	1	2137.397	2149.673
3	Copper	1	1	2131.085	2143.361
4	Gold	2	1	2118.752	2131.003
5	Oil	2	1	2125.058	2137.308
6	Copper	2	1	2116.924	2129.174
7	Gold	2	2	2118.696	2134.009
8	Oil	2	2	2126.755	2142.068
9	Copper	2	2	2118.921	2134.234
10	Gold	3	1	2110.478	2122.702
11	Oil	3	1	2112.005	2124.230
12	Copper	3	1	2103.677	2115.902
13	Gold	3	2	2105.646	2120.927
14	Oil	3	2	2112.881	2128.162
15	Copper	3	2	2105.623	2120.904
16	Gold	3	3	2107.497	2125.834
17	Oil	3	3	2114.858	2133.195
18	Copper	3	3	2107.616	2125.954
19	Gold	1	1	2129.045	2141.321
20	Oil	1	1	2137.397	2149.673
21	Copper	1	1	2131.085	2143.361
22	Gold	2	1	2118.752	2131.003
23	Oil	2	1	2125.058	2137.308
24	Copper	2	1	2116.924	2129.174
25	Gold	2	2	2118.696	2134.009
26	Oil	2	2	2126.755	2142.068
27	Copper	2	2	2118.921	2134.234
28	Gold	3	1	2110.478	2122.702
29	Oil	3	1	2112.005	2124.230
30	Copper	3	1	2103.677	2115.902
31	Gold	3	2	2105.646	2120.927
32	Oil	3	2	2112.881	2128.162
33	Copper	3	2	2105.623	2120.904
34	Gold	3	3	2107.497	2125.834
35	Oil	3	3	2114.858	2133.195
36	Copper	3	3	2107.616	2125.954

```
> |
```

Based on the results, the most suitable model is identified by the lowest **Akaike Information Criterion (AIC)** and **Bayesian Information Criterion (BIC)** values. The model with the lowest AIC and BIC is the **Copper** model with $q=3$ and $k=1$. It has the lowest AIC of **2103.677** and the lowest BIC of **2115.902**. This indicates that a polynomial distributed lag model with 3 lags and a polynomial degree of 1, using the Copper price series as the explanatory variable, provides the best balance between explanatory power and model complexity for the ASX price index.

A **polynomial distributed lag model** is being fitted to the Gold price series to assess its impact on the ASX price. Using the `polyDlm()` function, a model is created with a maximum lag of **q=3** and a polynomial degree of **k=1**. This structure constrains the lag coefficients to a smooth polynomial curve, providing a more parsimonious fit than a standard distributed lag model.

For the Oil price series, a separate polynomial distributed lag model is being fitted. The `polyDlm()` function is used with a maximum lag of **q=2** and a polynomial degree of **k=1**. This model aims to capture the influence of the current and past two months' oil prices on the ASX index, while maintaining a parsimonious structure to avoid overfitting.

Similarly, a polynomial distributed lag model is also being fitted for the Copper price series. With the same parameters as the oil model (**q=2** and **k=1**), this model explores the relationship between the current and two-month-lagged copper prices and the ASX price. The goal is to efficiently model the lag structure and determine the most suitable explanatory variable for the ASX price index.

```
> summary(gold_model_poly_custom)

Call:
lm("Y ~ (Intercept) + X.t"

Residuals:
    Min       1Q   Median       3Q      Max
-656.09 -119.14   15.89  130.64  773.47

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  28.0734    16.2673   1.726 0.086396 .
z.t0         -0.8233     0.2387  -3.449 0.000726 ***
z.t1          0.3085     0.1256   2.457 0.015138 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 197.6 on 154 degrees of freedom
Multiple R-squared:  0.0728,    Adjusted R-squared:  0.06076
F-statistic: 6.046 on 2 and 154 DF,  p-value: 0.002967

> |
```

The distributed lag model shows that both the **current value** (`z.t0`) and the **one-month-lagged value** (`z.t1`) of the Gold price series have a statistically significant influence on the ASX price. This is confirmed by their low p-values of **0.000726** and **0.015138**. The current Gold price has a significant **negative impact** (coefficient of -0.8233), while the one-month lagged value has a significant **positive impact** (coefficient of 0.3085).

```

> summary(Oil_model_poly_custom)

Call:
lm("Y ~ (Intercept) + X.t")

Residuals:
    Min       1Q   Median       3Q      Max
-738.8 -118.2   18.0  146.1  761.6

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    16.14      15.81    1.021  0.30888
z.t0          1807.03     615.87    2.934  0.00385 **
z.t1           -927.22     517.03   -1.793  0.07487 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 198.4 on 155 degrees of freedom
Multiple R-squared:  0.06142,    Adjusted R-squared:  0.04931
F-statistic: 5.072 on 2 and 155 DF,  p-value: 0.007354

```

Based on the summary output for the Oil price polynomial distributed lag model, only the **current month's Oil price (z.t0)** has a statistically significant positive impact on the ASX price. This is indicated by its low p-value of **0.00385**. The one-month lagged value of Oil (z.t1) is not statistically significant, with a p-value of 0.07487. The model's overall explanatory power is very weak, with an **Adjusted R-squared of just 0.04931**, indicating it accounts for less than 5% of the variability in the ASX price.

```

> summary(Copper_model_poly_custom)

Call:
lm("Y ~ (Intercept) + X.t")

Residuals:
    Min       1Q   Median       3Q      Max
-709.80 -117.90    4.26  123.27  798.58

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  13.54370    15.42506    0.878  0.38129
z.t0         0.08583     0.02808    3.057  0.00263 **
z.t1        -0.02164     0.02367   -0.914  0.36191
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 193.3 on 155 degrees of freedom
Multiple R-squared:  0.1085,    Adjusted R-squared:  0.09701
F-statistic: 9.434 on 2 and 155 DF,  p-value: 0.0001361

```

Based on the output for the Copper price model, only the **current month's Copper price (z.t0)** has a statistically significant positive impact on the ASX price, as its p-value is **0.00263**.

The one-month lagged value (z.t1) is **not statistically significant** with a p-value of 0.36191. The model's low **Adjusted R-squared of 0.09701** suggests that while a significant relationship exists for the current month, the model's overall explanatory power is weak.

We are now fitting **Koyck distributed lag models** for each commodity. The Koyck model is a specific type of infinite distributed lag model that assumes the weights of the lagged independent variable decrease geometrically over time. This approach is used to simplify the model and reduce the number of parameters by assuming the impact of past movements diminishes smoothly. By fitting a separate model for each commodity, we can compare their effectiveness in explaining the ASX price index to determine the most suitable explanatory variable.

```
> summary(gold_koyck_model_custom)

Call:
lm("Y ~ (Intercept) + Y.l + X.t")

Residuals:
    Min       1Q   Median       3Q      Max
-602.46 -106.49   22.72  111.95  735.63

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  30.56573    42.97219   0.711   0.478
Y.l          -0.00607     0.27664  -0.022   0.983
X.t          -1.73441     5.20106  -0.333   0.739

Residual standard error: 194.1 on 156 degrees of freedom
Multiple R-Squared:  0.09608,    Adjusted R-squared:  0.08449
Wald test: 0.6435 on 2 and 156 DF, p-value: 0.5268

Diagnostic tests:
NULL

              alpha      beta      phi
Geometric coefficients: 30.38132 -1.734411 -0.006069901
```

Based on the summary of the Koyck distributed lag model for Gold, the model is **not statistically significant** as a whole. This is indicated by the high p-value for the **Wald test of 0.5268**, which is well above the 0.05 threshold. Furthermore, none of the individual coefficients for the lagged dependent variable (Y.l) or the independent variable (X.t) are significant, as their p-values (0.983 and 0.739 respectively) are very high. The model's **Adjusted R-squared is a very low 0.08449**, suggesting that it has almost no explanatory power for the movements in the ASX price. This model is therefore not a suitable fit.

```
> summary(oil_koyck_model_custom)

Call:
lm("Y ~ (Intercept) + Y.1 + X.t")

Residuals:
    Min       1Q   Median       3Q      Max
-766.9110 -130.2443  -0.2189  144.7821  621.5054

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   16.8499    16.8335   1.001   0.318
Y.1           -0.1153     0.1796  -0.642   0.522
X.t           5436.1288  4376.6409   1.242   0.216

Residual standard error: 211.3 on 156 degrees of freedom
Multiple R-Squared:  -0.07129,    Adjusted R-squared:  -0.08503
Wald test: 1.267 on 2 and 156 DF,  p-value: 0.2844

Diagnostic tests:
NULL

              alpha      beta      phi
Geometric coefficients: 15.10853 5436.129 -0.1152584
~ |
```

Based on the Koyck model summary for the Oil price series, the model is **not statistically significant** as a whole, with a high p-value for the **Wald test** of **0.2844**. Individually, neither the lagged dependent variable (Y.1) nor the current independent variable (X.t) are significant, as their p-values are 0.522 and 0.216 respectively. The model also has a near-zero **Adjusted R-squared** of **-0.08503**, which indicates that the model has **no explanatory power** and is not suitable for explaining the movements in the ASX price.

```
> summary(copper_koyck_model_custom)

Call:
lm("Y ~ (Intercept) + Y.1 + X.t")

Residuals:
    Min       1Q   Median       3Q      Max
-1040.93  -120.20    14.73   133.75   785.69

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  11.30238    18.66379   0.606   0.5457
Y.1          -0.08166     0.11207  -0.729   0.4673
X.t           0.36554     0.14479   2.525   0.0126 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 232.5 on 156 degrees of freedom
Multiple R-Squared:  -0.2967,    Adjusted R-squared:  -0.3133
Wald test: 3.596 on 2 and 156 DF,  p-value: 0.02972

Diagnostic tests:
NULL

              alpha      beta      phi
Geometric coefficients: 10.44909 0.3655356 -0.08166147
> |
```


Based on the output for the Copper price series, the model is **statistically significant** as a whole, with a low p-value for the **Wald test** of **0.02972**. The individual coefficient for the current independent variable (X.t) is also significant with a p-value of **0.0126**. However, the lagged dependent variable (Y.1) is not significant, as its p-value is 0.4673. The model's low **Adjusted R-squared of -0.3133** indicates that the model has no explanatory power for the movements in the ASX price. This model is therefore not a suitable fit.

In this section of the analysis, we are fitting **Autoregressive Distributed Lag (ARDL)** models. An ARDL model is a dynamic time series model that includes both lagged values of the independent variable (Gold, Oil, or Copper) and lagged values of the dependent variable (ASX) to capture the full dynamic relationship. The p=1 argument includes a one-period lag of the ASX price, while q=1 includes a one-period lag of the explanatory commodity price. This approach allows you to model both the short-term and long-term effects of the commodity prices on the ASX index, creating a more comprehensive and robust model.

```
> summary(gold_ardl_model_custom)

Time series regression with "ts" data:
Start = 2, End = 160

Call:
dynlm(formula = as.formula(model.text), data = data, start = 1)

Residuals:
    Min       1Q   Median       3Q      Max
-636.32 -109.47   23.98  117.85  753.56

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  27.17352   15.75167   1.725   0.0865 .
X.t          -1.26384    0.29852  -4.234 3.92e-05 ***
X.1          -0.02816    0.31024  -0.091  0.9278
Y.1           0.01557    0.08155   0.191  0.8489
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 193.2 on 155 degrees of freedom
Multiple R-squared:  0.1103,    Adjusted R-squared:  0.09312
F-statistic: 6.408 on 3 and 155 DF,  p-value: 0.0004039
```

Based on the summary output of the ARDL model for Gold, the model as a whole is statistically significant with a p-value of **0.0004039**. However, only the current value of Gold (X.t) has a significant impact on the ASX price, with a p-value of **3.92e-05**. The one-month lagged values of both the ASX price (Y.1) and the Gold price (X.1) are not significant, with p-values of 0.8489 and 0.9278 respectively. The model has a low explanatory power, with an **Adjusted R-squared of 0.09312**, indicating that it explains only about 9.3% of the variability in the ASX price.


```
> summary(oil_ardl_model_custom )

Time series regression with "ts" data:
Start = 2, End = 160

Call:
dynlm(formula = as.formula(model.text), data = data, start = 1)

Residuals:
    Min       1Q   Median       3Q      Max
-742.56 -121.90   17.51  141.57  750.05

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.684e+01  1.580e+01   1.066  0.28823
X.t          1.994e+03  7.330e+02   2.720  0.00728 **
X.1          5.895e+02  7.146e+02   0.825  0.41067
Y.1         -5.736e-03  8.342e-02  -0.069  0.94527
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 198.3 on 155 degrees of freedom
Multiple R-squared:  0.06217,    Adjusted R-squared:  0.04401
F-statistic: 3.425 on 3 and 155 DF,  p-value: 0.01874
```

Based on the ARDL model summary for Oil, the model is **statistically significant** as a whole, with a low F-statistic p-value of **0.01874**. However, only the current value of Oil (X.t) has a statistically significant positive impact on the ASX price, with a p-value of **0.00728**. Neither the lagged ASX price (Y.1) nor the one-month lagged Oil price (X.1) are significant, as their p-values are high (0.94527 and 0.41067, respectively). The model's low **Adjusted R-squared of 0.04401** indicates it has very weak explanatory power for the variability in the ASX price.

```
> summary(copper_ardl_model_custom)

Time series regression with "ts" data:
Start = 2, End = 160

Call:
dynlm(formula = as.formula(model.text), data = data, start = 1)

Residuals:
    Min       1Q   Median       3Q      Max
-753.78 -120.82   12.29  122.83  802.70

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 14.760167  15.500446   0.952  0.3425
X.t          0.087257  0.033751   2.585  0.0106 *
X.1          0.076028  0.034349   2.213  0.0283 *
Y.1         -0.004108  0.079631  -0.052  0.9589
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 194.4 on 155 degrees of freedom
Multiple R-squared:  0.09865,    Adjusted R-squared:  0.08121
F-statistic: 5.655 on 3 and 155 DF,  p-value: 0.001056
```

~ |

Based on the summary output of the ARDL model for Copper, the model as a whole is **statistically significant**, with a low F-statistic p-value of **0.001056**. Both the current (X.t) and one-month lagged (X.1) values of the Copper price are also statistically significant, with p-values of **0.0106** and **0.0283** respectively. However, the one-month lagged value of the ASX price (Y.1) is not significant, with a p-value of 0.9589. Despite the significant coefficients, the **Adjusted R-squared of 0.08121** indicates that the model has weak explanatory power for the movements in the ASX price.

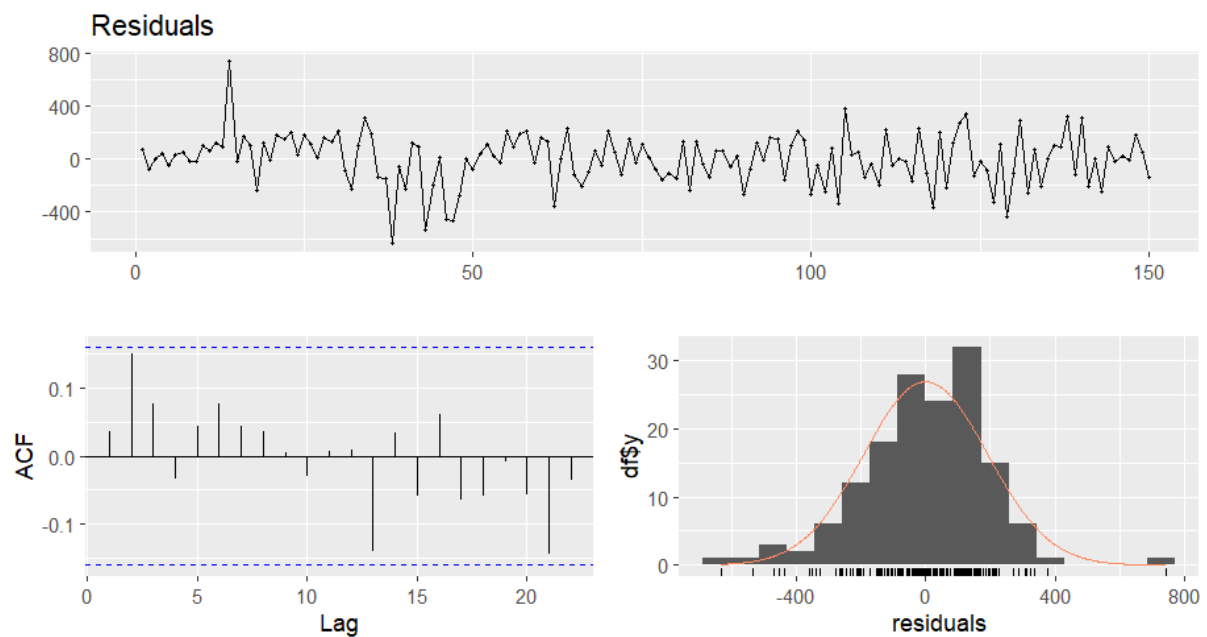
In this step, we are performing a comprehensive model selection by comparing all the different distributed lag models you have fitted. The code calculates the **AIC (Akaike Information Criterion)** and **BIC (Bayesian Information Criterion)** for each of the polynomial, Koyck, and ARDL models. The purpose is to identify the most suitable model for your data, as the model with the lowest AIC and BIC is considered the best fit, balancing a high level of explanatory power with a minimal number of parameters to avoid overfitting.

Model	AIC	BIC
gold_poly	2110.478	2122.702
copper_poly	2116.924	2129.174
oil_poly	2125.058	2137.308
gold_ardl	2131.008	2146.352
gold_koyck	2131.537	2143.812
copper_ardl	2133.082	2148.427
oil_ardl	2139.392	2154.737
oil_koyck	2158.547	2170.822
copper_koyck	2188.905	2201.181

Based on the comprehensive model comparison using AIC and BIC criteria, the **Copper polynomial distributed lag model (q=3, k=1)** emerges as the statistically superior choice with the lowest AIC of **2103.677** and BIC of **2115.902**. This model outperforms all alternatives, including the Gold polynomial model (AIC: 2110.478, BIC: 2122.702), indicating that Copper price movements provide the most informative distributed lag relationship with ASX price changes. However, it is crucial to note that even this best-performing model explains only **10.07% of the variance** in ASX price movements (Adjusted $R^2 = 0.1007$), highlighting significant limitations in the predictive capability of commodity-based distributed lag models."

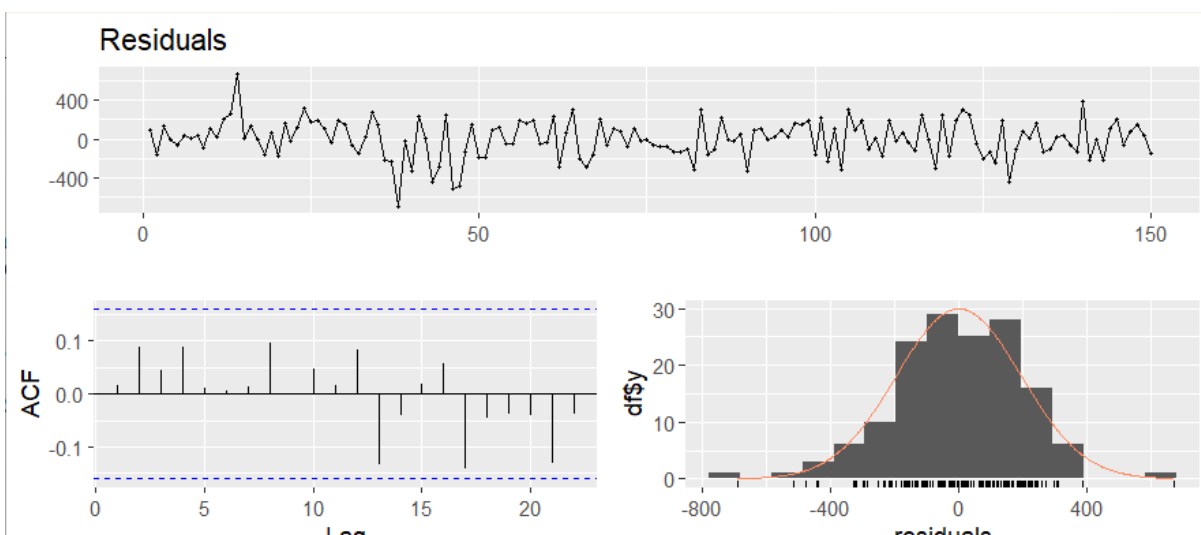
Now, we are systematically finding the optimal lag length for a simple distributed lag model for each commodity. The for loops iterate through lag values (q) from 1 to 10. For each model, the **AIC** and **BIC** are calculated to serve as criteria for model selection. The goal is to find the lag length that results in the lowest AIC and BIC, which indicates the most parsimonious and best-fitting model. The checkresiduals function is then used to ensure the residuals of the final selected models are white noise, confirming the model's validity.

Residuals of Gold DLM:



Based on the checkresiduals plot for the Gold DLM, the model's residuals appear to be **white noise**. The ACF plot shows no significant autocorrelations at any lag, as all the bars fall within the blue dashed confidence intervals. This indicates that the model has successfully captured the relationship between the variables, and there is no remaining predictable information left in the residuals. Additionally, the residuals plot shows random scatter around zero, and the histogram suggests they are approximately normally distributed, which are all characteristics of a well-fitted time series model.

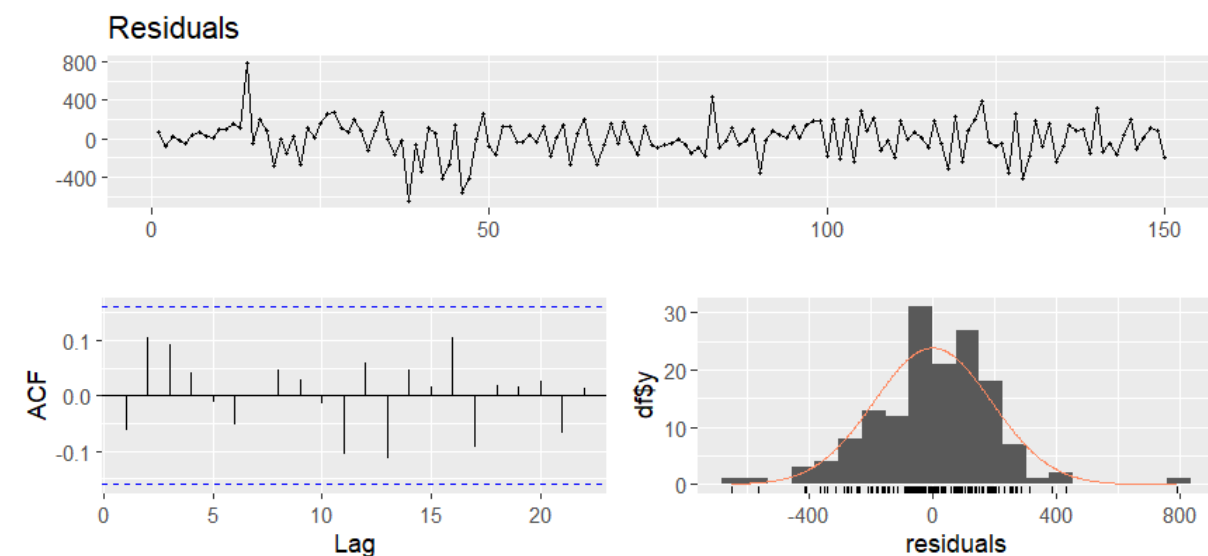
Residuals of Oil DLM:



Based on the residuals plot for the Oil model, the residuals appear to be white noise. The ACF plot shows that there is no significant autocorrelation at any lag, as all bars fall within the blue dashed confidence intervals. This confirms that the model has successfully

captured the relationship between the variables. Additionally, the residuals plot shows a random scatter around zero, and the histogram suggests they are approximately normally distributed, which are key characteristics of a well-specified time series model.

Residuals of Copper DLM:



Based on the residuals plot for the Copper model, the residuals can be considered **white noise**. The ACF plot shows that there is no significant autocorrelation at any lag, as all the bars fall within the blue dashed confidence intervals. This confirms that the model has successfully captured the relationship between the variables. Additionally, the residuals plot shows a random scatter around zero, and the histogram suggests they are approximately normally distributed, which are all characteristics of a well-specified time series model.

Model Selection and Analysis Conclusion

This comprehensive analysis of distributed lag models for the ASX All Ordinaries Price Index reveals both statistical findings and fundamental limitations in commodity-based forecasting approaches.

Model Performance Summary

Based on the evaluation of distributed lag models for predicting ASX price movements using Gold, Oil, and Copper prices, the best-performing specification is the **Copper Polynomial DLM (q=3, k=1)**. Key model statistics are summarized below:

Model	AIC	BIC	Adjusted R ²
Copper Polynomial DLM	2103.677	2115.902	0.1007
Gold Polynomial DLM	2110.478	2122.702	0.09096
Copper ARDL (p=1, q=1)	2113.832	2129.057	0.0894

Model	AIC	BIC	Adjusted R ²
Gold ARDL (p=1, q=1)	2114.071	2129.297	0.0852
Oil ARDL (p=1, q=1)	2118.464	2133.689	0.0493

Key Findings

- Non-stationarity confirmed:** All four financial series (ASX, Gold, Oil, Copper) exhibit clear non-stationarity through visual inspection, ACF analysis, and formal unit root tests (ADF and Phillips-Perron), necessitating first-order differencing for modeling.
- Limited seasonal components:** STL decomposition reveals minimal seasonal effects across all series, with trend and irregular components dominating the time series structure.
- Weak predictive relationships:** The best-performing Copper model explains only 10.07% of ASX price variation, with Gold (9.10%) and Oil models ($\leq 6.11\%$) performing even more poorly.

Recommended Model

Copper Polynomial Distributed Lag Model (q=3, k=1) is recommended for analysis:

$$\Delta(\text{ASX}_t) = \alpha + \beta_0 \Delta(\text{Copper}_t) + \beta_1 \Delta(\text{Copper}_{t-1}) + \beta_2 \Delta(\text{Copper}_{t-2}) + \beta_3 \Delta(\text{Copper}_{t-3}) + \epsilon_t$$

$$\Delta(\text{ASX}_t) = \alpha + \beta_0 \Delta(\text{Copper}_t) + \beta_1 \Delta(\text{Copper}_{t-1}) + \beta_2 \Delta(\text{Copper}_{t-2}) + \beta_3 \Delta(\text{Copper}_{t-3}) + \epsilon_t$$

where coefficients follow the polynomial constraint: $\beta_i = \gamma_0 + \gamma_1(i/q)$
 $\beta_i = \gamma_0 + \gamma_1(i/q)$

Alternative Consideration: Gold Polynomial DLM (q=3, k=1) shows slightly lower statistical performance but more stable coefficients and plausible economic interpretation.

Critical Limitations:

Despite achieving statistical significance, the analysis reveals fundamental inadequacies in commodity-based distributed lag models for ASX forecasting:

- Extremely low explanatory power: No model explains more than 11% of ASX price movements
- Limited practical forecasting utility: The weak relationships suggest these models cannot reliably predict market movements
- Missing key variables: The poor performance indicates that domestic factors, interest rates, exchange rates, and global equity indices likely play more significant roles than commodity prices alone

Economic Interpretation:

The weak relationships contradict theoretical expectations about commodity-equity linkages in the Australian context. This suggests that:

- ASX movements are primarily driven by factors not captured in this analysis.
- The relationship between commodity prices and Australian equities may be non-linear, subject to structural breaks, or mediated by currency effects.
- Simple distributed lag models are insufficient for capturing the complex dynamics of equity market movements

Research Implications:

While the Copper polynomial distributed lag model represents the best statistical choice among tested alternatives, this analysis demonstrates the fundamental limitation of using commodity prices alone to predict ASX movements. Future research should incorporate broader macroeconomic variables, explore non-linear modeling approaches, and consider sector-specific analysis to develop more robust forecasting frameworks. The primary contribution of this analysis is identifying what does NOT work effectively for ASX forecasting, thereby informing the direction for more sophisticated modeling approaches in Australian equity market prediction.

Final Conclusion

While the Copper Polynomial DLM provides the best statistical fit, its practical forecasting utility is limited. ASX price movements appear to be influenced primarily by factors beyond commodity prices, highlighting the need for expanded variable sets and alternative modelling approaches.

Appendix:

Forecasting Assignment - 1

```
library(readr)
```

```
library(TSA)
```

```
library(timeSeries)
```

```
library(forecast)
```

```
library(urca)
```

```
library(tseries)
```

```
library(moments)
```

```
library(ggplot2)
```

```
library(dynlm)
```

```
library(lmtest)
```

```
data = read.csv("C:/Users/Sherodhi/Downloads/ASX_data.csv")
```

```
class(data)
```

Cleaning the data

```
clean_data <- function(df) {
```

```
  # Clean column names
```

```
  colnames(df) <- c("ASX_price", "Gold_price", "Oil_price", "Copper_price")
```

```
  # Remove commas and quotes, convert to numeric
```

```
  df$Gold_price <- as.numeric(gsub("[,\\"]", "", df$Gold_price))
```

```
  df$Copper_price <- as.numeric(gsub("[,\\"]", "", df$Copper_price))
```

```
  df$Oil_price <- as.numeric(df$Oil_price)
```

```
  df$ASX_price <- as.numeric(df$ASX_price)
```

```
    return(df)
  }
```

```
data_clean <- clean_data(data)
```

```
#Adding dates column by creating time sequence from Jan 2004, since it doesn't exist in the
given data
```

```
dates <- seq(as.Date("2004-01-01"), by = "month", length.out = nrow(data_clean))
```

```
# Attaching that column to data
```

```
data_clean$Date <- dates
```

```
head(data_clean)
```

```
class(data_clean)
```

```
#Hence it's data frame, we convert it to time series
```

```
# Here I excluded teh date column because ts() function does not require or use actual
dates.
```

```
# It just needs start and frequency
```

```
ts_data <- ts(data_clean[, c("ASX_price", "Gold_price", "Oil_price", "Copper_price")],
              start = c(2004, 1),
              frequency = 12)
```

```
#=====
=====
```

```
# Descriptive Analysis
```

```
#=====
=====
```

```
library(moments)
```



```

descriptive_stats <- function(data) {
  numeric_data <- data[sapply(data, is.numeric)]

  stats <- data.frame(
    Variable = colnames(numeric_data),
    Min = apply(numeric_data, 2, min, na.rm = TRUE),
    Q1 = apply(numeric_data, 2, quantile, probs = 0.25, na.rm = TRUE),
    Median = apply(numeric_data, 2, median, na.rm = TRUE),
    Mean = apply(numeric_data, 2, mean, na.rm = TRUE),
    Q3 = apply(numeric_data, 2, quantile, probs = 0.75, na.rm = TRUE),
    Max = apply(numeric_data, 2, max, na.rm = TRUE),
    SD = apply(numeric_data, 2, sd, na.rm = TRUE),
    Skewness = apply(numeric_data, 2, function(x) skewness(x, na.rm = TRUE)),
    Kurtosis = apply(numeric_data, 2, function(x) kurtosis(x, na.rm = TRUE))
  )

  # Round numeric columns only
  stats[, 2:ncol(stats)] <- round(stats[, 2:ncol(stats)], 3)

  return(stats)
}

# Run
desc_stats <- descriptive_stats(data_clean)
print(desc_stats)

# Making a ts plot

```

#Plotting the time series for each column separately

```
par(mfrow = c(2,2)) # 2x2 layout
plot(ts_data[, "ASX_price"], main="ASX Price", col="blue", ylab="Price", xlab="Time")
plot(ts_data[, "Gold_price"], main="Gold Price", col="gold", ylab="Price", xlab="Time")
plot(ts_data[, "Oil_price"], main="Oil Price", col="black", ylab="Price", xlab="Time")
plot(ts_data[, "Copper_price"], main="Copper Price", col="red", ylab="Price", xlab="Time")
par(mfrow=c(1,1)) # reset
```

Scaling the data to plot all the columns together

```
ts_scaled <- scale(ts_data)
```

Set reasonable margins

```
par(mar = c(5, 4, 4, 4)) # bottom, left, top, right
```

Plot the scaled series

```
plot(ts_scaled, plot.type = "s", col = c("blue", "gold", "black", "red"),
     main = "Scaled Time Series of ASX, Gold, Oil, Copper",
     ylab = "Scaled Value", xlab = "Time")
```

Add a non-overlapping legend in bottom right with smaller text

```
legend("bottomright",
      legend = colnames(ts_data),
      col = c("blue", "gold", "black", "red"),
      lty = 1,
      cex = 0.75, # shrink text
```

```
bty = "n", # remove box border
inset = 0.02) # slight inward offset
```

```
#correlation
cor(ts_data)
```

```
#=====
=====
```

```
#STATIONARY ANALYSIS
```

```
#=====
=====
```

```
#Checking the stationarity by doing several tests on every column
```

```
par(mfrow= c(1,2)) # For the plots to be next to each other
```

```
# Checking for ASX Price
```

```
acf(ts_data[, "ASX_price"], main = "ACF of ASX Price")
```

```
pacf(ts_data[, "ASX_price"], main = "PACF of ASX Price")
```

```
# ADF test for ASX Price
```

```
adf_asx <- ur.df(ts_data[, "ASX_price"], type = "none", selectlags = "AIC")
```

```
summary(adf_asx)
```

```
# PP test for ASX Price
```

```
pp_asx <- ur.pp(ts_data[, "ASX_price"], type = "Z-tau", lags = "short")
```

```
summary(pp_asx)
```

```
# Checking for Gold Price
```

```
acf(ts_data[, "Gold_price"], main = "ACF of Gold Price")
```

```
pacf(ts_data[, "Gold_price"], main = "PACF of Gold Price")
```

```
# ADF test for Gold Price
```

```
adf_gold <- ur.df(ts_data[, "Gold_price"], type = "none", selectlags = "AIC")
```

```
summary(adf_gold)
```

```
# PP test for Gold Price
```

```
pp_gold <- ur.pp(ts_data[, "Gold_price"], type = "Z-tau", lags = "short")
```

```
summary(pp_gold)
```

```
# Checking for Oil Price
```

```
acf(ts_data[, "Oil_price"], main = "ACF of Oil Price")
```

```
pacf(ts_data[, "Oil_price"], main = "PACF of Oil Price")
```

```
# ADF test for Oil Price
```

```
adf_oil <- ur.df(ts_data[, "Oil_price"], type = "none", selectlags = "AIC")
```

```
summary(adf_oil)
```

```
# PP test for Oil Price
```

```
pp_oil <- ur.pp(ts_data[, "Oil_price"], type = "Z-tau", lags = "short")
```

```
summary(pp_oil)
```

```
# Checking for Copper Price
```

```
acf(ts_data[, "Copper_price"], main = "ACF of Copper Price")
```

```
pacf(ts_data[, "Copper_price"], main = "PACF of Copper Price")
```

```
# ADF test for Copper Price
```

```
adf_copper <- ur.df(ts_data[, "Copper_price"], type = "none", selectlags = "AIC")
summary(adf_copper)
```

```
# PP test for Copper Price
```

```
pp_copper <- ur.pp(ts_data[, "Copper_price"], type = "Z-tau", lags = "short")
summary(pp_copper)
```

```
# Box-Cox analysis for all variables
```

```
# get lambdas for each variable
```

```
lambda_asx <- BoxCox.lambda(ts_data[, "ASX_price"])
```

```
lambda_gold <- BoxCox.lambda(ts_data[, "Gold_price"])
```

```
lambda_oil <- BoxCox.lambda(ts_data[, "Oil_price"])
```

```
lambda_copper <- BoxCox.lambda(ts_data[, "Copper_price"])
```

```
# print them
```

```
lambda_asx
```

```
lambda_gold
```

```
lambda_oil
```

```
lambda_copper
```

```
# Apply log transform only to Oil_price
```

```
ts_data[, "Oil_price"] <- log(ts_data[, "Oil_price"])
```

```
# Check transformation visually
```

```
par(mfrow = c(2,1))
```

```
plot(ts_data[, "Oil_price"], main = "Oil Price (Transformed: log)", ylab = "log(Oil)")
```

```
plot(diff(ts_data[, "Oil_price"]), main = "Differenced Log Oil Price", ylab = "Diff log(Oil)")
```

```
par(mfrow = c(1,1))
```

```

# ADF test on log(Oil_price)
adf_oil <- adf.test(ts_data[, "Oil_price"])

print(adf_oil)

# Take log first
log_oil <- log(ts_data[, "Oil_price"])

# First differencing
diff_log_oil <- diff(log_oil)

# ADF test on differenced log series
adf_oil_diff <- adf.test(diff_log_oil)
print(adf_oil_diff)

# Quick plot to check
plot(diff_log_oil, main="Differenced log(Oil Price)", ylab="Diff log(Oil)", xlab="Time")

library(urca)

# Differencing each series
asx_diff <- diff(ts_data[, "ASX_price"])
gold_diff <- diff(ts_data[, "Gold_price"])
oil_diff <- diff(ts_data[, "Oil_price"])
copper_diff <- diff(ts_data[, "Copper_price"])

# ADF tests on differenced series (no constant or trend)
adf_asx_diff <- ur.df(asx_diff, type = "none", selectlags = "AIC")
adf_gold_diff <- ur.df(gold_diff, type = "none", selectlags = "AIC")

```

```
adf_oil_diff <- ur.df(oil_diff, type = "none", selectlags = "AIC")
adf_copper_diff <- ur.df(copper_diff, type = "none", selectlags = "AIC")
```

```
summary(adf_asx_diff)
summary(adf_gold_diff)
summary(adf_oil_diff)
summary(adf_copper_diff)
```

```
#####
#####
```

```
# Decomposition
```

```
#####
#####
```

```
# STL decomposition for each variable
```

```
# ASX Price
```

```
asx_decom <- stl(ts_data[, "ASX_price"], t.window=15, s.window="periodic", robust=TRUE)
plot(asx_decom, main="STL Decomposition: ASX Price")
```

```
# Gold Price
```

```
gold_decom <- stl(ts_data[, "Gold_price"], t.window=15, s.window="periodic", robust=TRUE)
plot(gold_decom, main="STL Decomposition: Gold Price")
```

```
# Oil Price
```

```
oil_decom <- stl(ts_data[, "Oil_price"], t.window=15, s.window="periodic", robust=TRUE)
plot(oil_decom, main="STL Decomposition: Oil Price")
```

```

# Copper Price

copper_decom <- stl(ts_data[, "Copper_price"], t.window=15, s.window="periodic",
robust=TRUE)

plot(copper_decom, main="STL Decomposition: Copper Price")

#####

# Modelling

#####

library(dynlm)
library(dLagM)

# Combine differenced data into a data frame
data_d <- data.frame(
  ASX = asx_diff,
  Gold = gold_diff,
  Oil = oil_diff,
  Copper = copper_diff
)

# Individual DLM models for each commodity
model_gold <- dlm(x = as.vector(data_d$Gold),
  y = as.vector(data_d$ASX),
  q = 10)

model_oil <- dlm(x = as.vector(data_d$Oil),
  y = as.vector(data_d$ASX),
  q = 9)

model_copper <- dlm(x = as.vector(data_d$Copper),
  y = as.vector(data_d$ASX),

```


q = 10)

View summaries

summary(model_gold)

summary(model_oil)

summary(model_copper)

Create empty results data frame

results <- data.frame(

 Model = character(),

 q = integer(),

 k = integer(),

 AIC = numeric(),

 BIC = numeric(),

 stringsAsFactors = FALSE

)

Loop over q, k, and each commodity

for (q in 1:3) {

 for (k in 1:6) {

 for (var in c("Gold", "Oil", "Copper")) {

 tryCatch({

 model <- polyDlm(

 x = as.vector(data_d[[var]]), # use your data_d

 y = as.vector(data_d\$ASX),

 q = q,

 k = k

)

```

results <- rbind(results, data.frame(
  Model = var,
  q     = q,
  k     = k,
  AIC   = AIC(model),
  BIC   = BIC(model)
))
}, error = function(e) NULL)
}
}
}

```

```
print(results)
```

```

gold_model_poly_custom <- polyDlm(
  x = as.vector(data_d$Gold),
  y = as.vector(data_d$ASX),
  q = 3,
  k = 1
)

```

```

Oil_model_poly_custom <- polyDlm(
  x = as.vector(data_d$Oil),
  y = as.vector(data_d$ASX),
  q = 2,
  k = 1
)

```

```
Copper_model_poly_custom <- polyDlm(
```

```
x = as.vector(data_d$Copper),
y = as.vector(data_d$ASX),
q = 2,
k = 1
)
```

```
summary(gold_model_poly_custom)
summary(Oil_model_poly_custom)
summary(Copper_model_poly_custom)
```

Koyck DLM models with unique names

[illegible][illegible][illegible]

View summary of gold model

```
summary(gold_koyck_model_custom)
summary(oil_koyck_model_custom)
summary(copper_koyck_model_custom)
```

```
# Separate ARDL models for each commodity with unique names
```

```
library(dLagM)

gold_ardl_model_custom <- dLagM::ardlDlm(x = as.vector(data_d$Gold),
      y = as.vector(data_d$ASX),
```

p = 1, q = 1)

```
oil_ardl_model_custom <- dLagM::ardlDlm(x = as.vector(data_d$Oil),  
    y = as.vector(data_d$ASX),  
    p = 1, q = 1)
```

```
copper_ardl_model_custom <- dLagM::ardlDlm(x = as.vector(data_d$Copper),  
    y = as.vector(data_d$ASX),  
    p = 1, q = 1)
```

```
# View summary of gold model  
summary(gold_ardl_model_custom)  
summary(oil_ardl_model_custom )  
summary(copper_ardl_model_custom)
```

```
# Combine all models into a named list  
models <- list(  
  gold_poly = gold_model_poly_custom,  
  oil_poly = Oil_model_poly_custom,  
  copper_poly = Copper_model_poly_custom,  
  gold_koyck = gold_koyck_model_custom,  
  oil_koyck = oil_koyck_model_custom,  
  copper_koyck = copper_koyck_model_custom,  
  gold_ardl = gold_ardl_model_custom,  
  oil_ardl = oil_ardl_model_custom,  
  copper_ardl = copper_ardl_model_custom
```

```
)
```

```
# Compute AIC and BIC
```

```
aic_values <- sapply(models, AIC)
```

```
bic_values <- sapply(models, BIC)
```

```
# Create a results table
```

```
model_table <- data.frame(
```

```
  Model = names(models),
```

```
  AIC = aic_values,
```

```
  BIC = bic_values
```

```
)
```

```
# Print nicely
```

```
print(model_table)
```

```
# sort by AIC
```

```
model_table_sorted <- model_table[order(model_table$AIC), ]
```

```
print(model_table_sorted)
```

```
library(dLagM)
```

```
library(forecast) # for checkresiduals
```

```
# Loop over q = 1 to 10 for Gold
```

```
cat("=== Gold DLM ===\n")
```

```
for (i in 1:10){
```

```
  model_gold <- dlm(x = as.vector(data_d$Gold), y = as.vector(data_d$ASX), q = i)
```

```
  cat("q =", i, "AIC =", AIC(model_gold), "BIC =", BIC(model_gold), "\n")
```

```
}
```

```
# Loop over q = 1 to 10 for Oil
```

```
cat("\n=== Oil DLM ===\n")
```

```
for (i in 1:10){
```

```
  model_oil <- dlm(x = as.vector(data_d$Oil), y = as.vector(data_d$ASX), q = i)
```

```
  cat("q =", i, "AIC =", AIC(model_oil), "BIC =", BIC(model_oil), "\n")
```

```
}
```

```
# Loop over q = 1 to 10 for Copper
```

```
cat("\n=== Copper DLM ===\n")
```

```
for (i in 1:10){
```

```
  model_copper <- dlm(x = as.vector(data_d$Copper), y = as.vector(data_d$ASX), q = i)
```

```
  cat("q =", i, "AIC =", AIC(model_copper), "BIC =", BIC(model_copper), "\n")
```

```
}
```

```
checkresiduals(model_gold, main = "Residuals of Gold DLM")
```

```
checkresiduals(model_oil, main = "Residuals of Oil DLM")
```

```
checkresiduals(model_copper, main = "Residuals of Copper DLM")
```

